

Local Interconnect Network (LIN)

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Session Objectives

- To identify the significance of LIN protocol
- To understand the characteristics and operation of LIN protocol



Session Topics

- Introduction to LIN
- Characteristics of LIN Protocol
- LIN node and network
- LIN frame structure
- LIN frame format
- Network management of LIN cluster
- Transport layer in LIN
- Error handling
- LIN-PHY specification



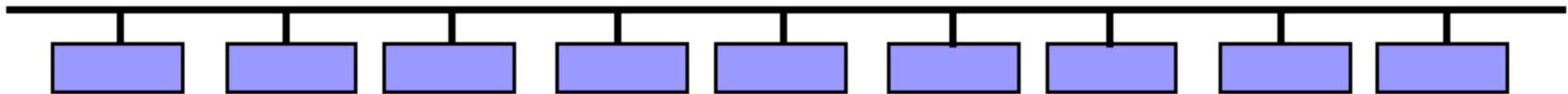
Introduction to LIN

- The primary and original purpose of LIN is to provide a 'sub-bus' for CAN, with reduced functionality and lower costs
- An economical solution when the requisite performance level is not high
- Can be used for applications where
 - bit rate requirements of the network are low
 - Reliability and robustness found in CAN are not required

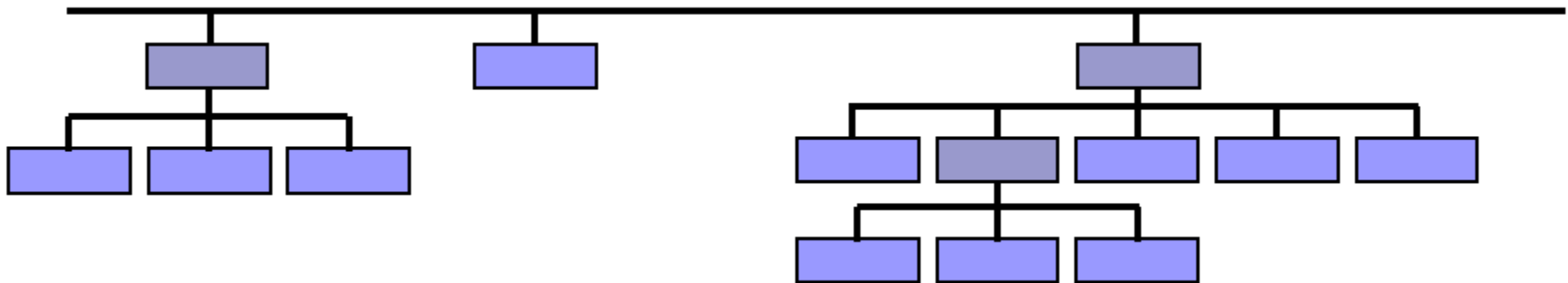


Hierarchical Network Structure

Flat Architecture



Hierarchical Architecture



- Hierarchical architecture reduces load on the main bus





Characteristics of LIN Protocol

- Mono-master, up to 16 slaves
- Single wire bus
- Bitrates from 1 to 20 Kbits/s : 2.4, 9.6 and 19.2 Kbits are usually used in automotive applications
- Multicast (broadcast) messages



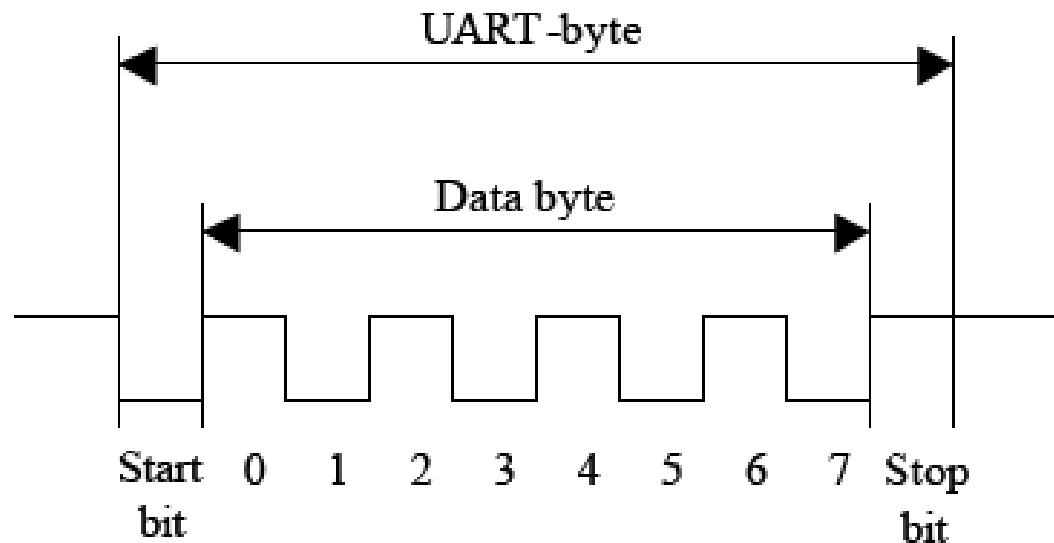
Characteristics of LIN Protocol

- Self-synchronization of the slave (only the master has an accurate clock as crystal)
- Messages with 2,4 or 8 data bytes, 3 control bytes
- Error detection by 8 bits checksum and 2 parity bits in identifier
- Sleep / wake-up capability

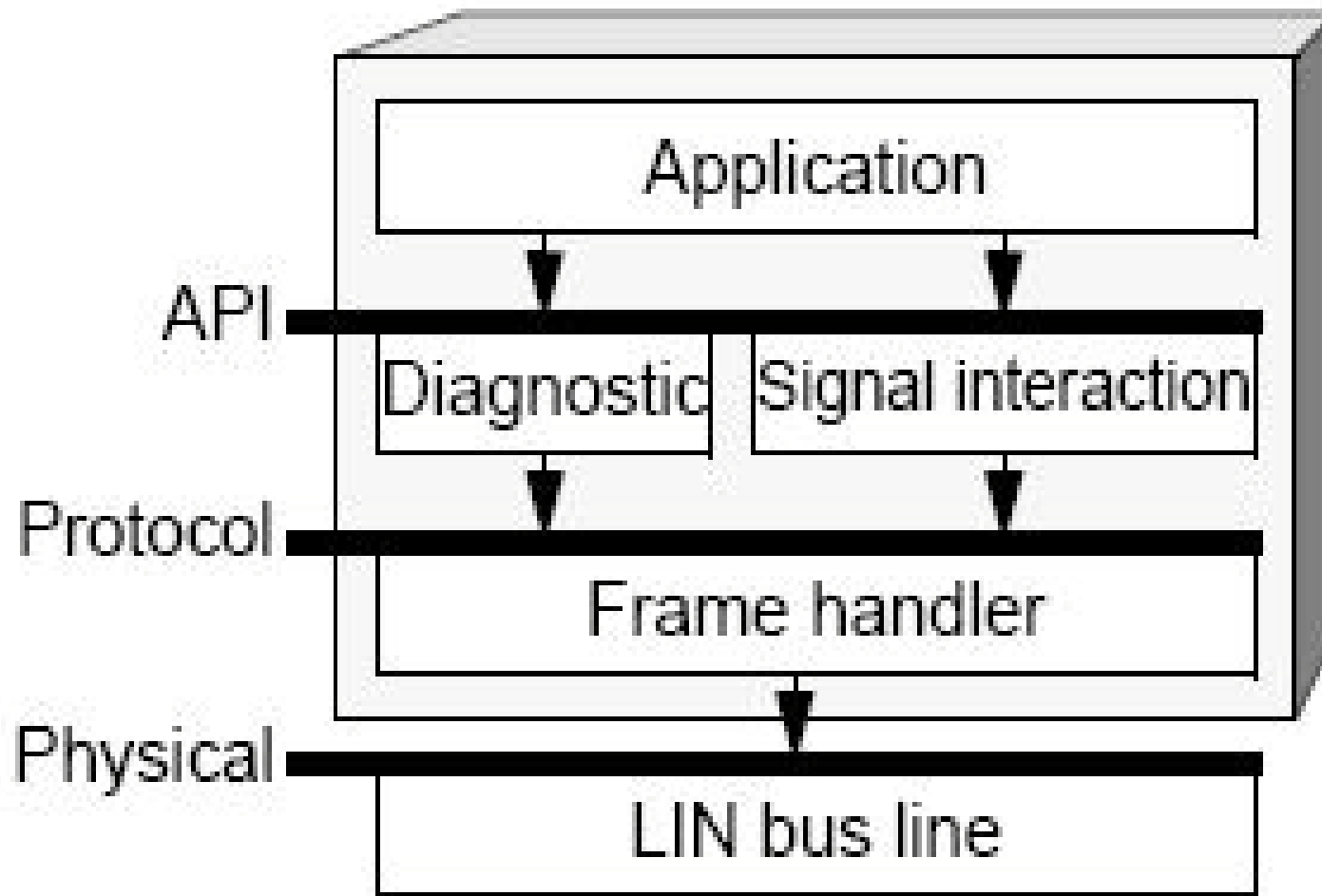


UART Bus

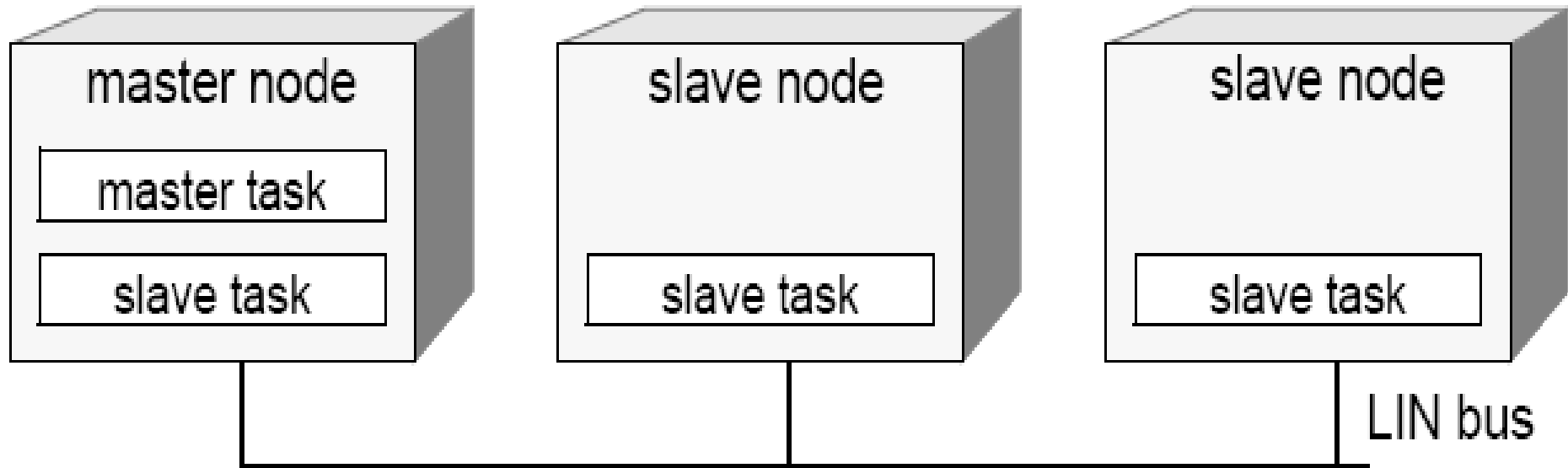
- Stands for Universal Asynchronous Rceiver / Transmitter
- All messages sent on the LIN-bus are byte-encoded according to the UART-protocol



LIN Node



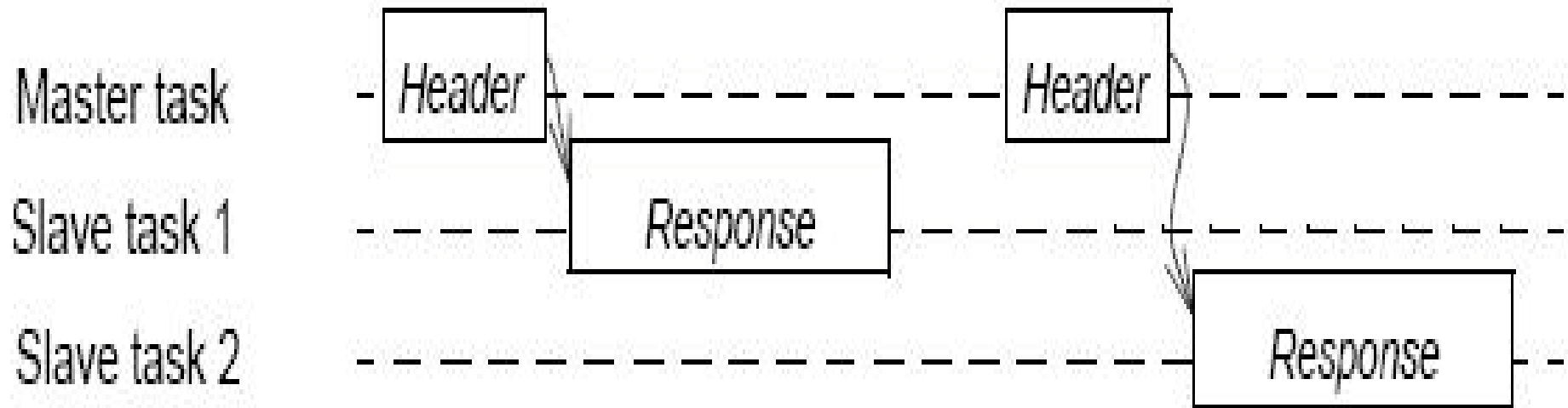
LIN Network



- Master task decides when and which frame shall be transferred on the bus
- Slave tasks provide the data transported by each frame



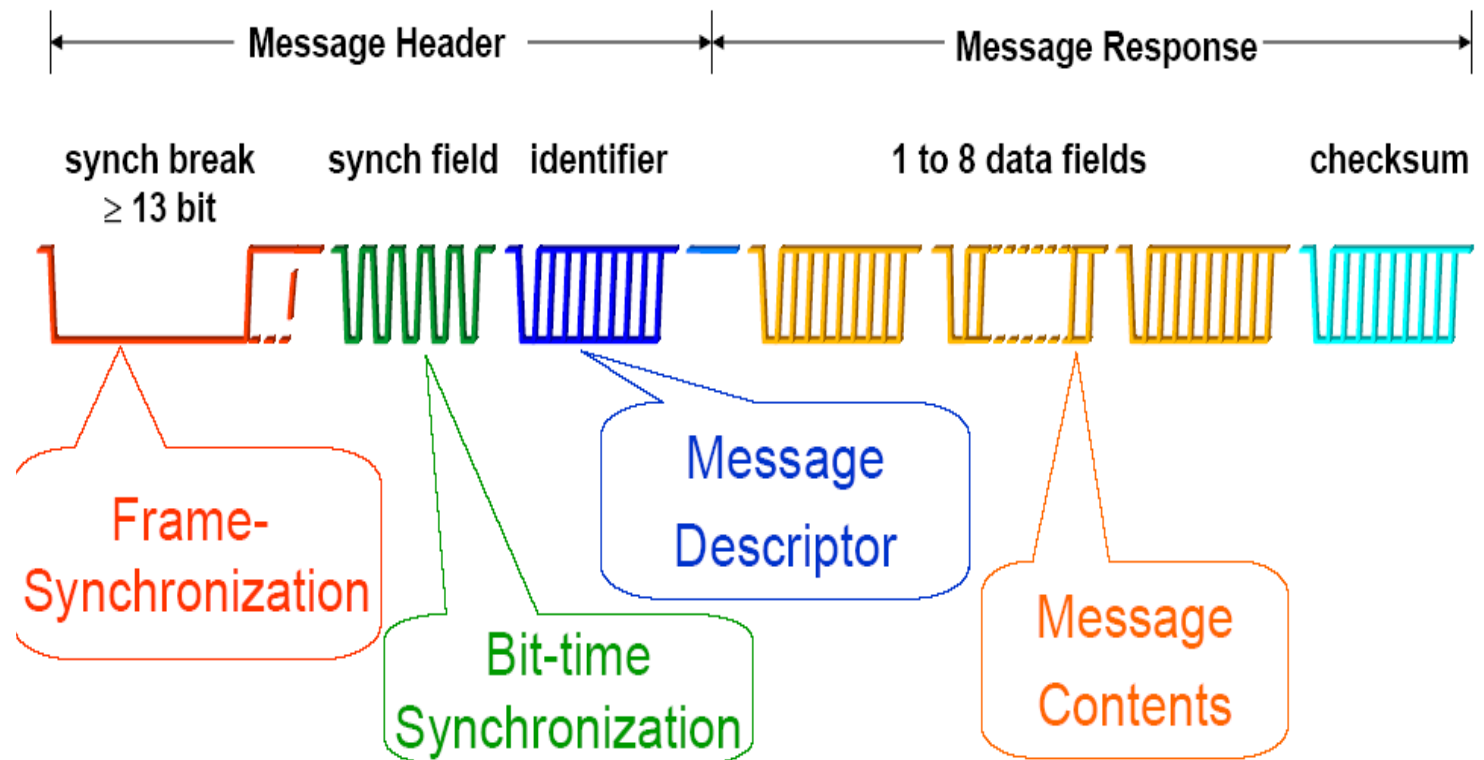
LIN Frame Structure



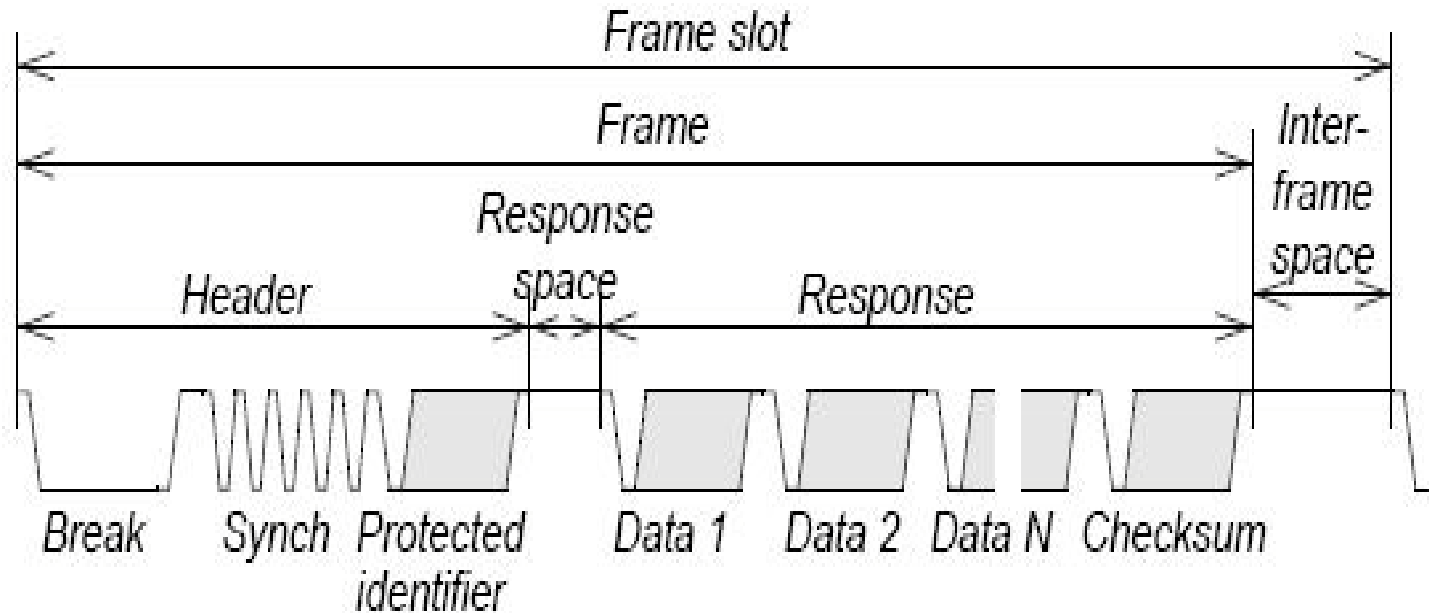
- A frame comprises of
 - Header and
 - Response

LIN Frame Structure

- Headers are situated in a master task comprises of
 - Breaks,
 - Synchronization pattern and
 - Identifier

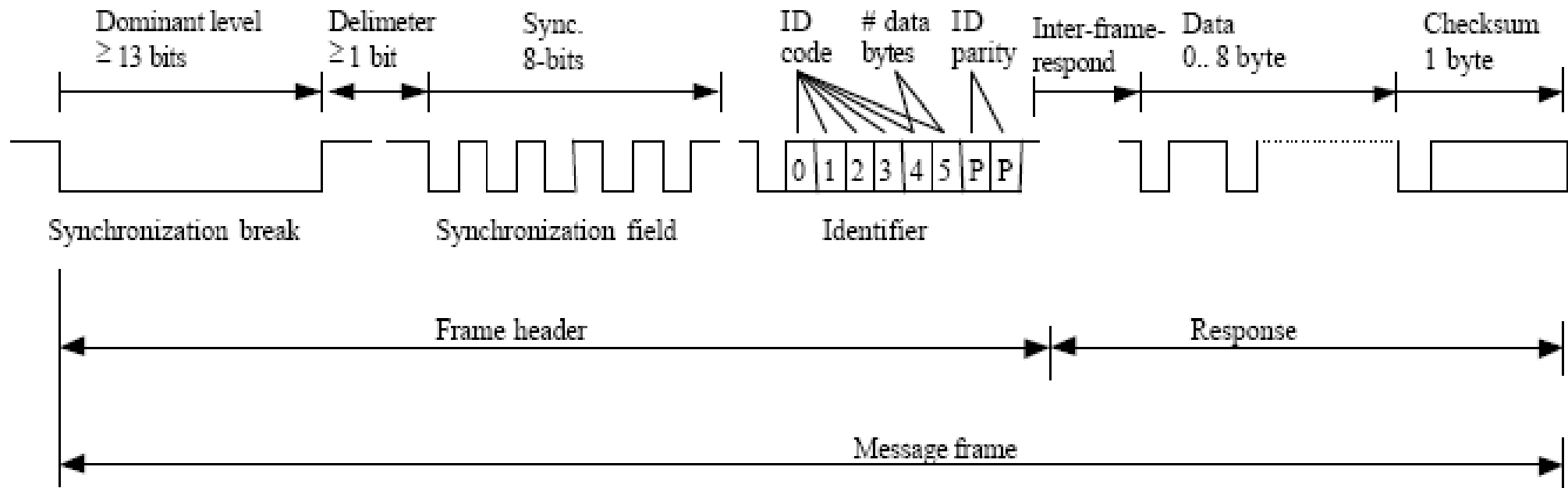


LIN Frame Format



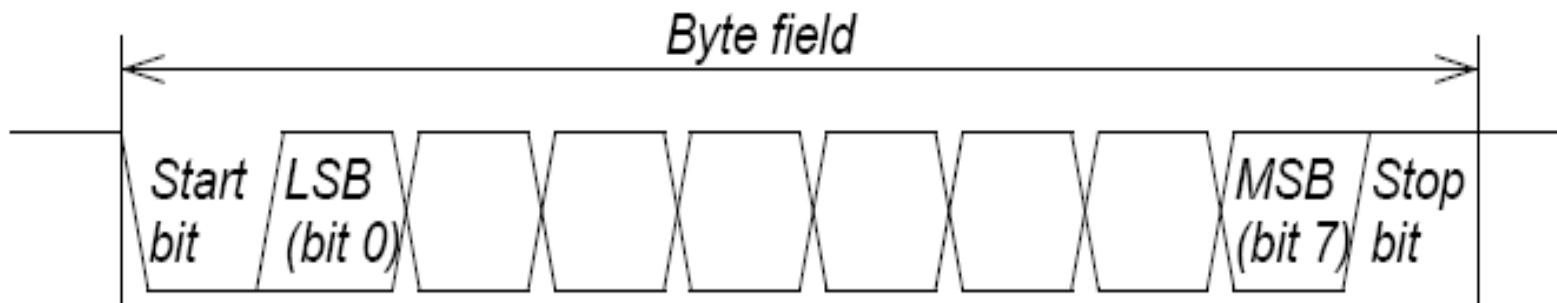
- Response space is the duration between transmission of header and response on the medium

LIN Frame Format



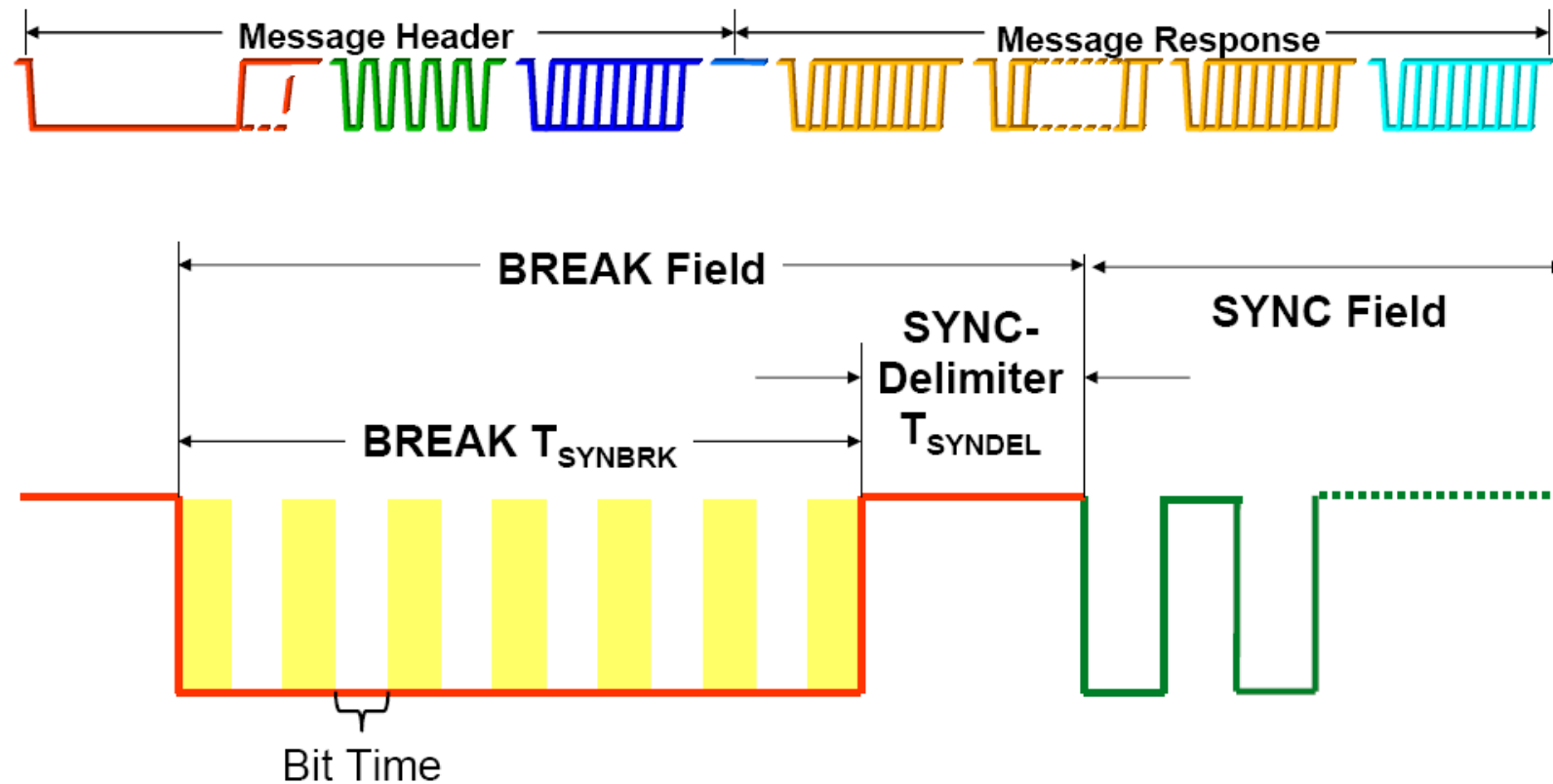
Byte Field in LIN

- Each byte field is transmitted as a serial byte
- The LSB of the data is sent first and the MSB last
- The start bit is encoded as a bit with value zero (dominant) and the stop bit is encoded as a bit with value one (recessive)



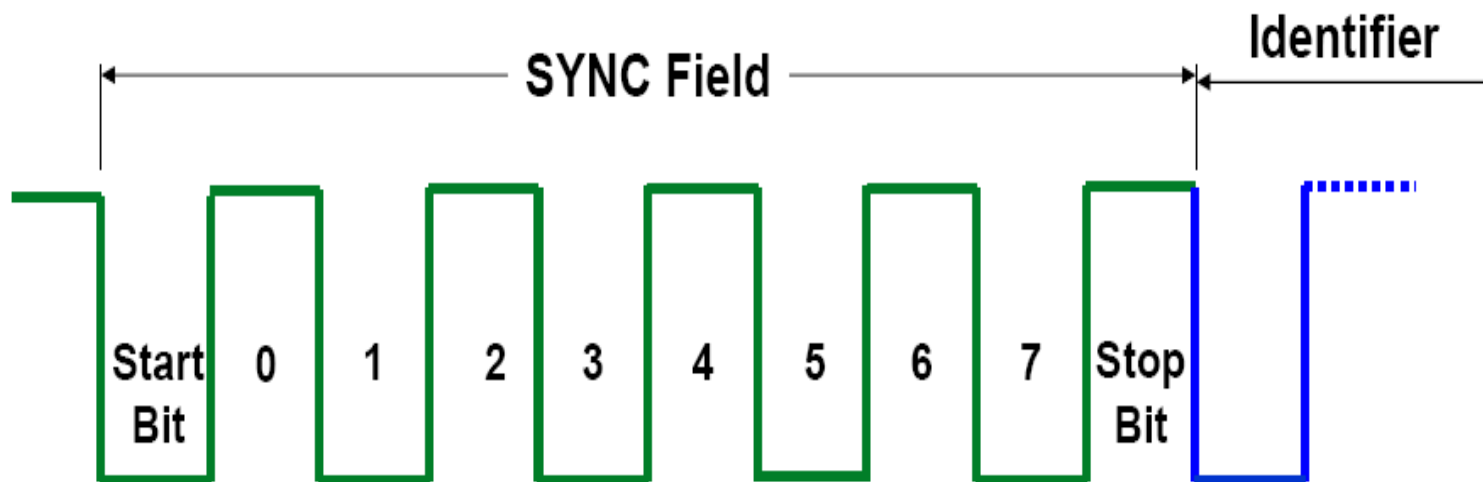
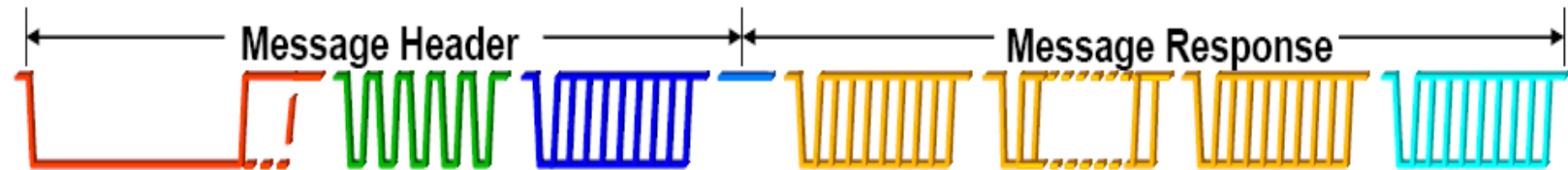
Break Field

- Signals the beginning of a new frame
- Always generated by the master task (in the master node)



Synchronization Field

- Initiated by master by sending a header
- Used by slaves to synchronize their clocks

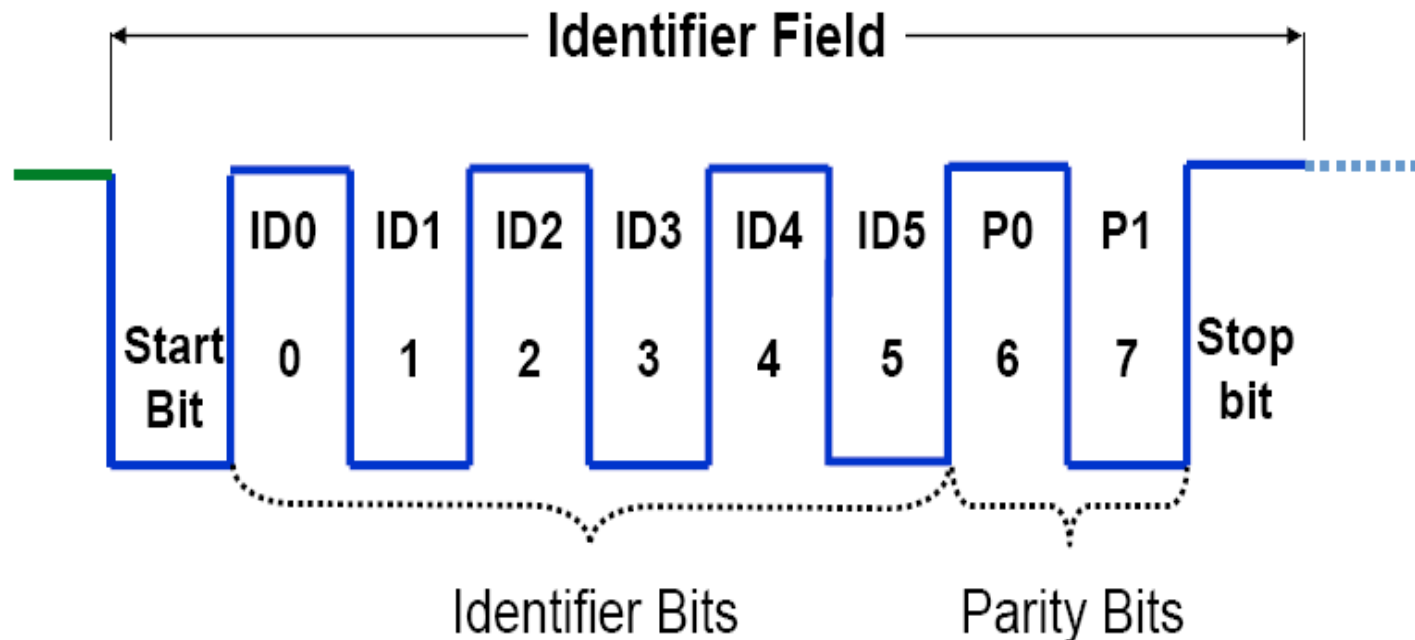
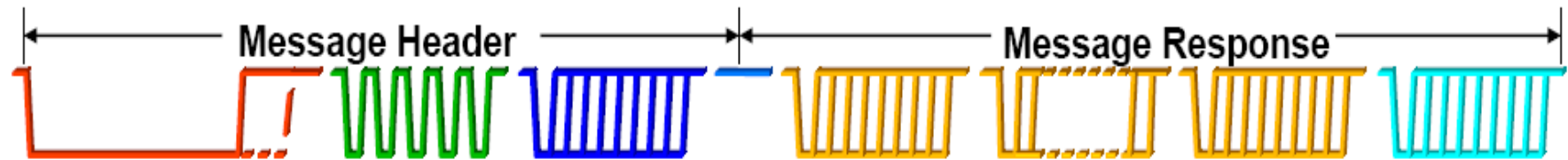


Protected Identifier Field

- Comprises of
 - Identifier
 - Identifier parity
- Six bits are reserved for the identifier (ID)
- The identifiers are split in four categories:
 - Values 0 to 59 (0x3b) are used for signal-carrying frames,
 - 60 (0x3c) and 61 (0x3d) are used to carry diagnostic data,
 - 62 (0x3e) is reserved for user-defined extensions,
 - 63 (0x3f) is reserved for future protocol enhancements

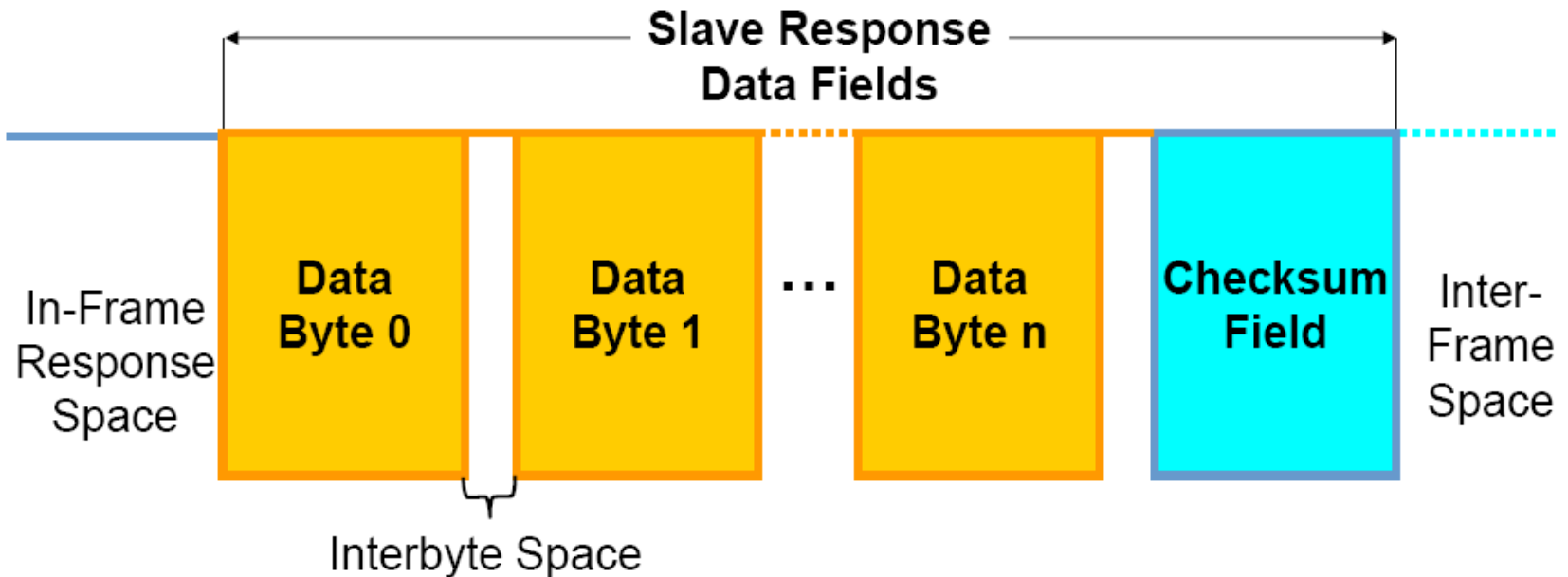
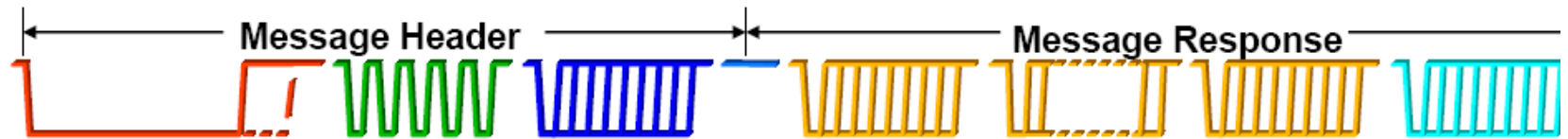


Protected Identifier Field

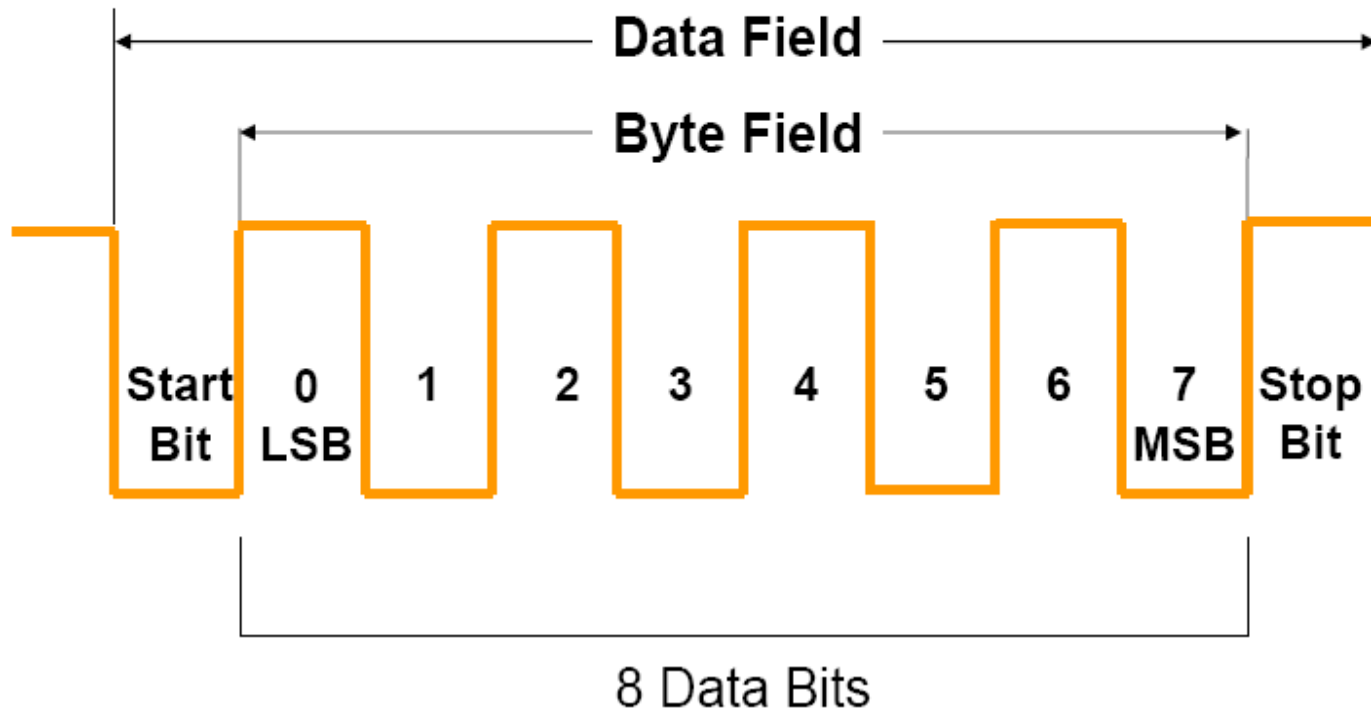
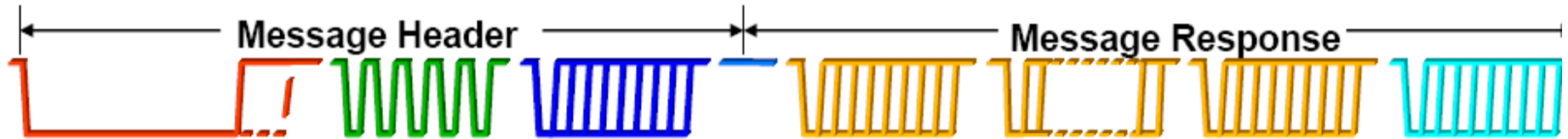


Data Field

- A frame carries between one and eight bytes of data
- The data fields are labeled data 1, data 2,... up to maximum data 8

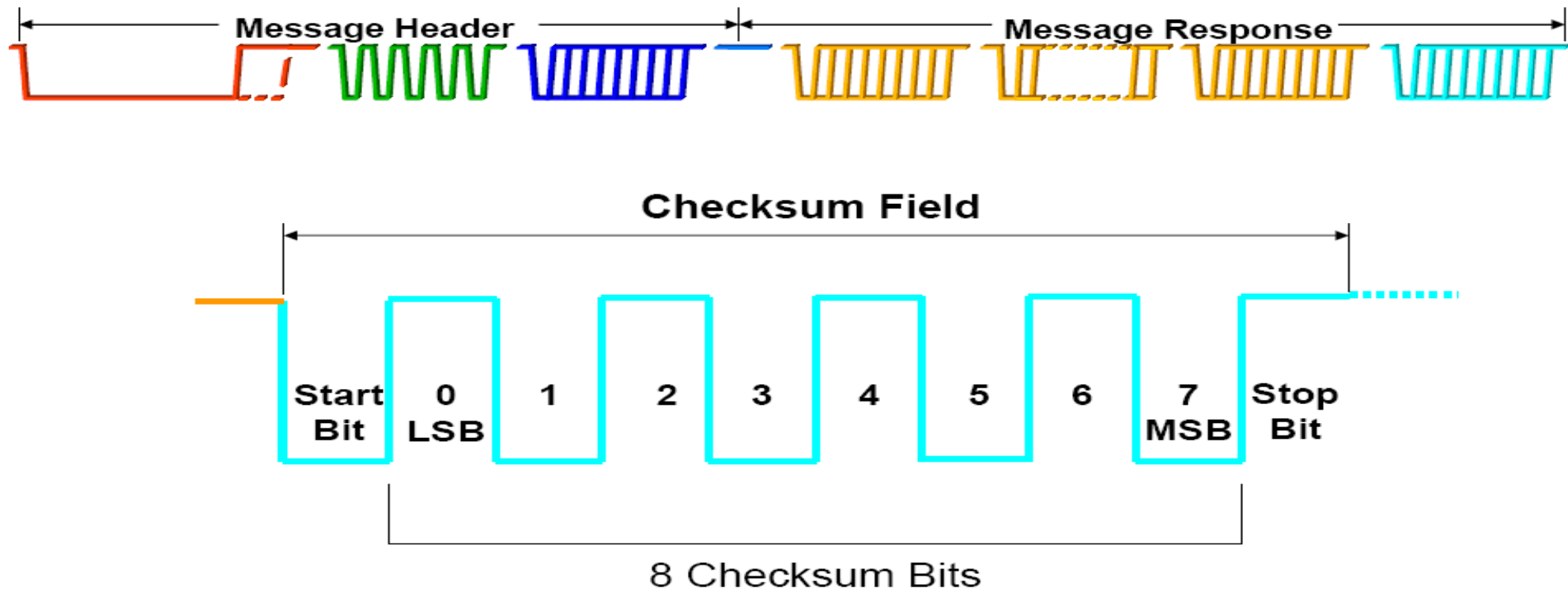


Data Byte

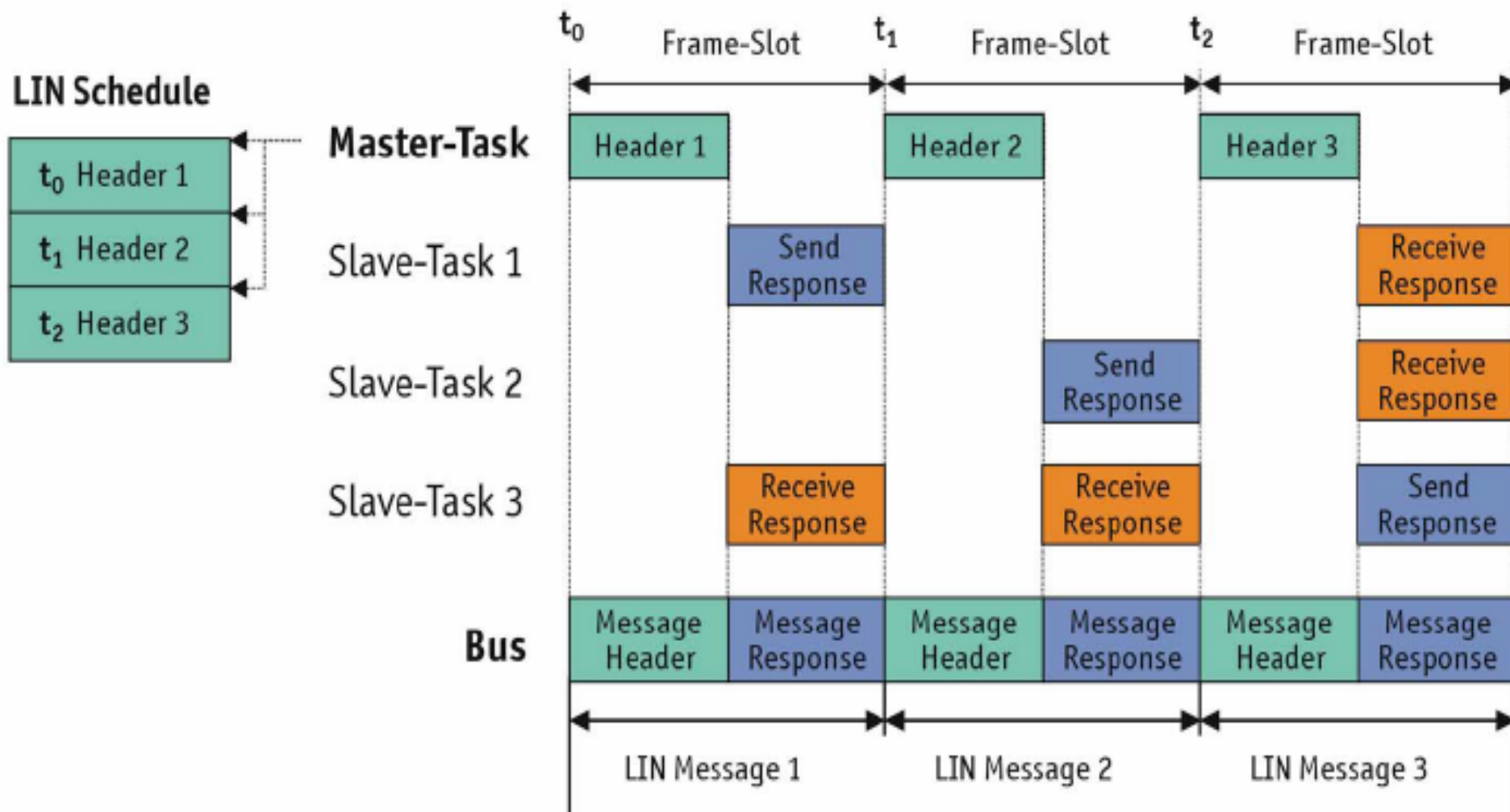


Checksum Field

- Contains the inverted eight bit sum with carry over all data bytes **or** all data bytes and the protected identifier

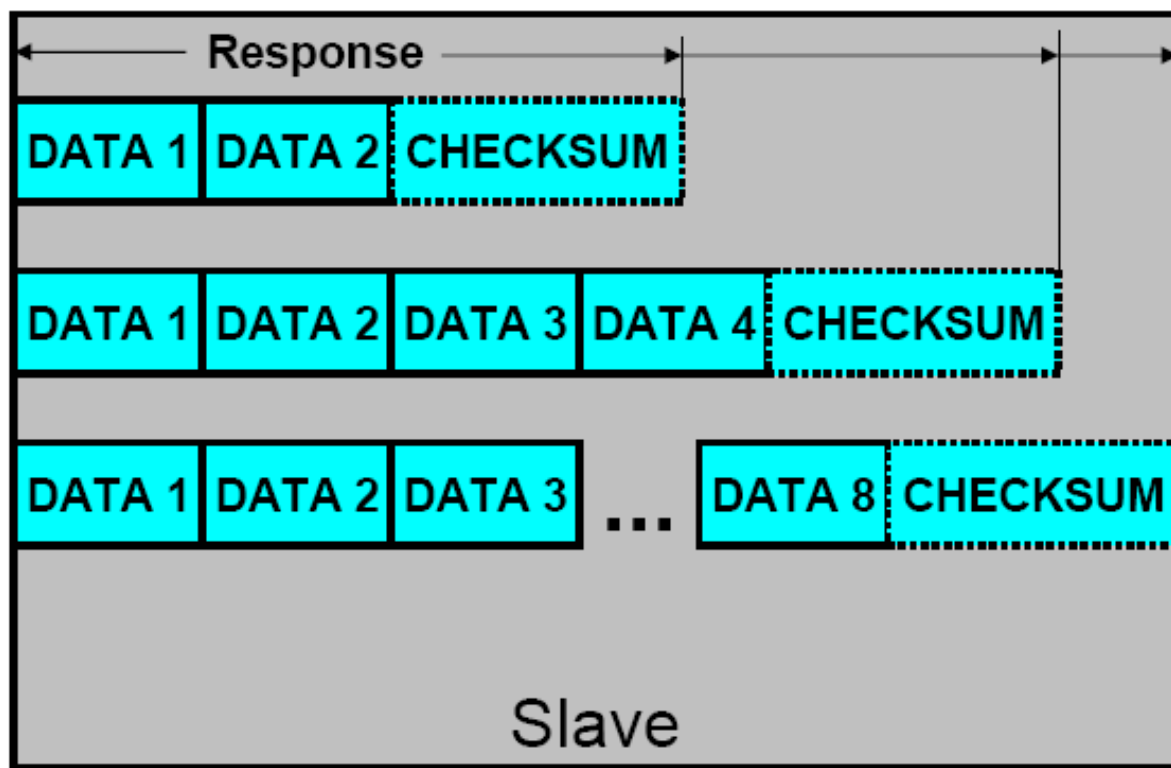
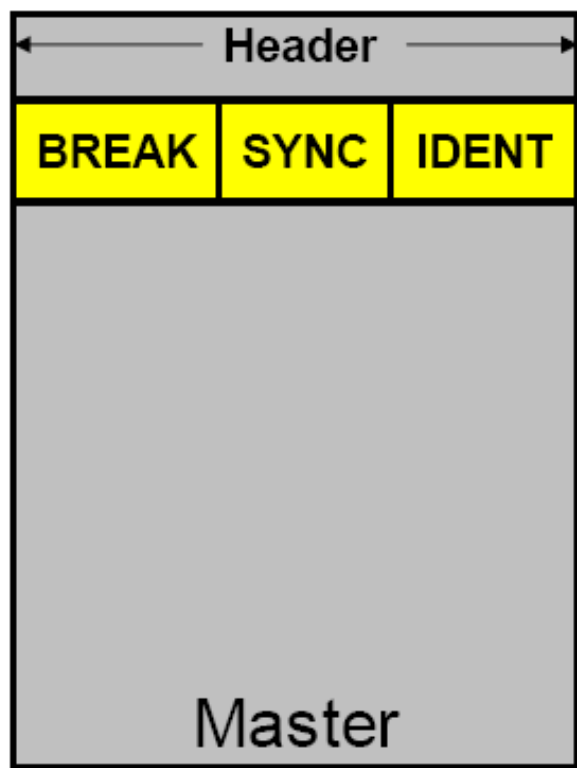


Frame Exchange over LIN Network



Frame Exchange over LIN Network

In-Frame Response Space

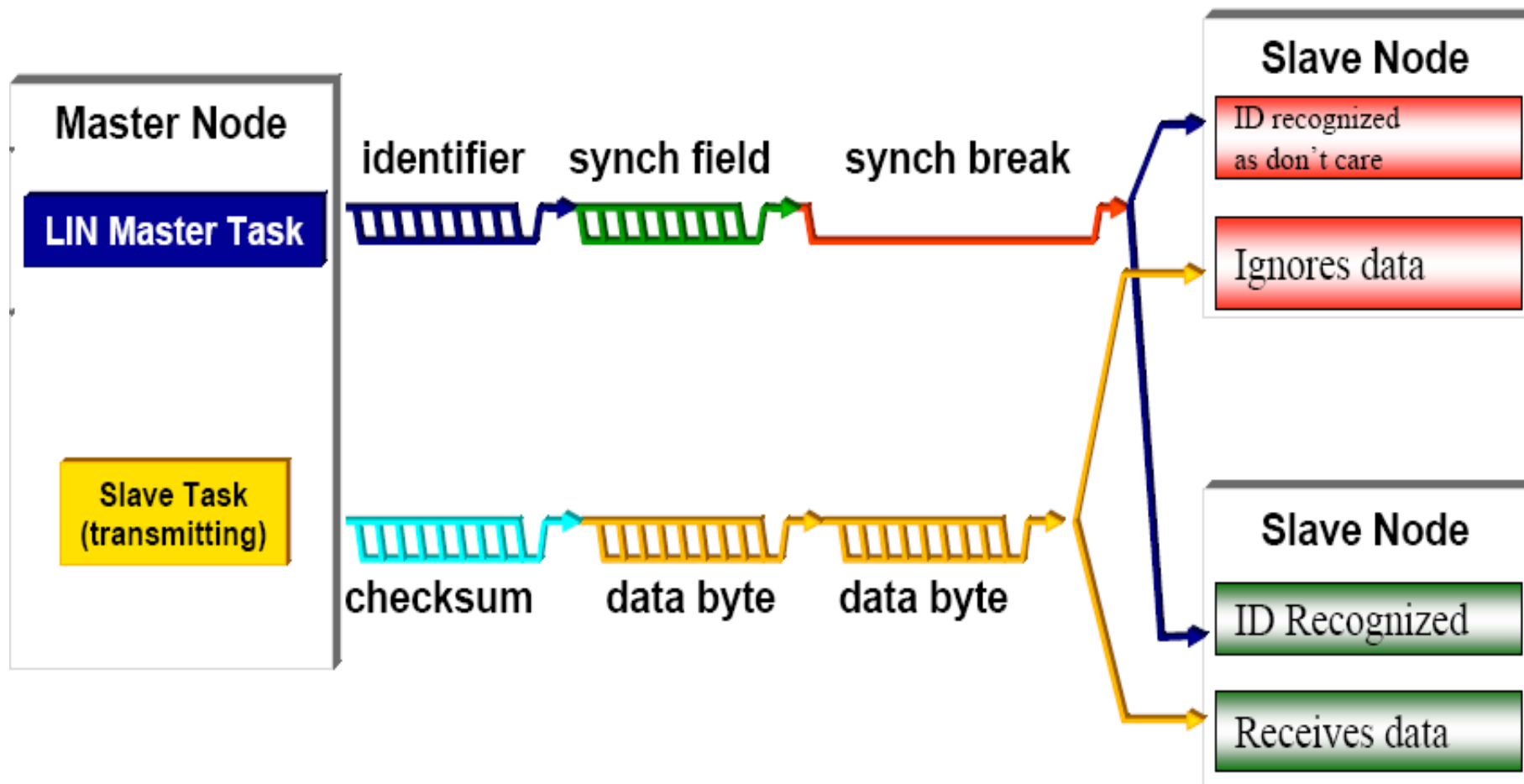


Frame Exchange over LIN Network

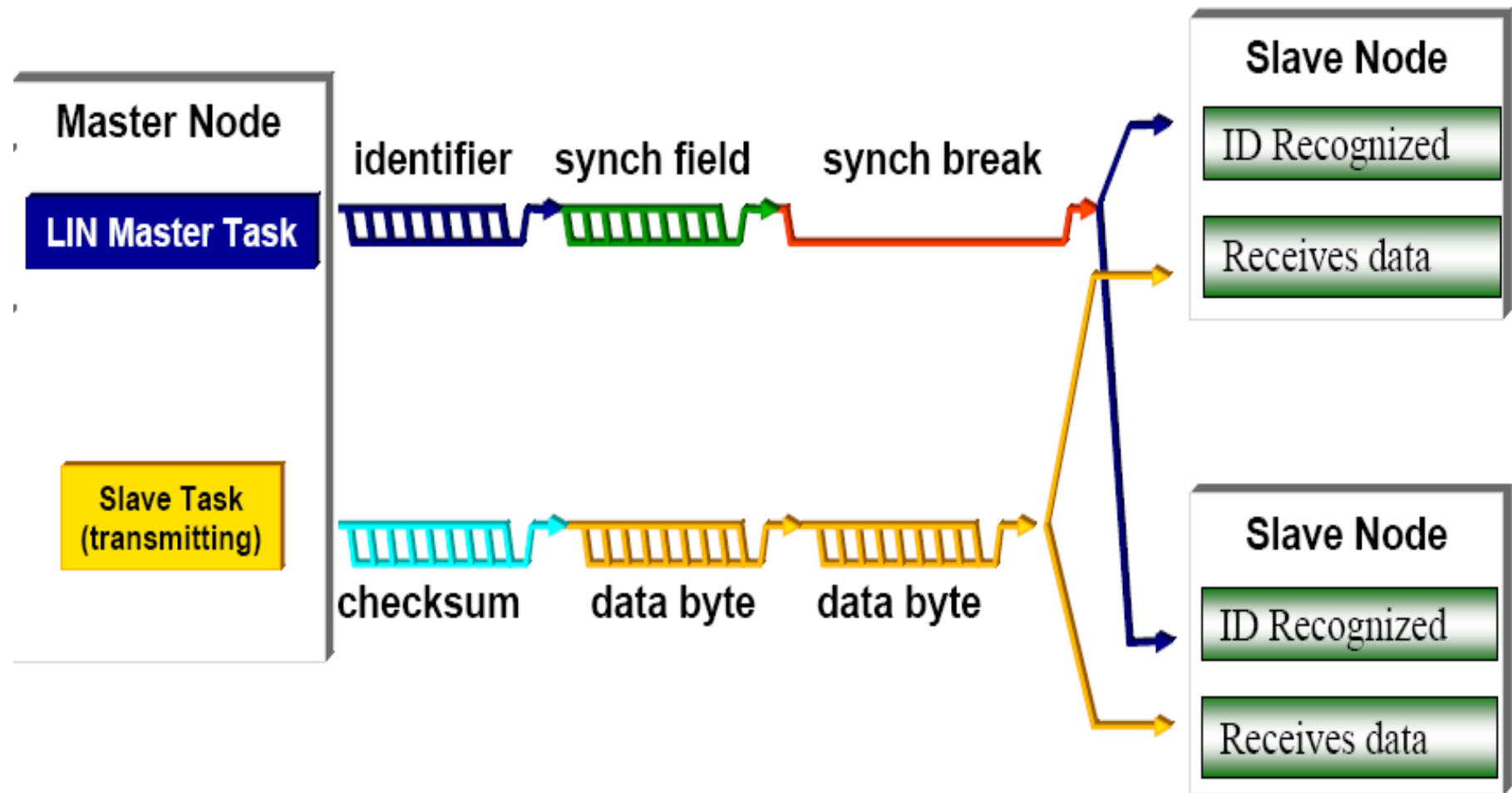
- The flow of information in a LIN network occurs from Master to Slave. There can be the following possibilities:
 - Master sends data to one slave (Unicast)
 - Master sends data to many slaves (Multicast)
 - Master Polls Slaves for Data
 - Data transfer from Slave to Slave



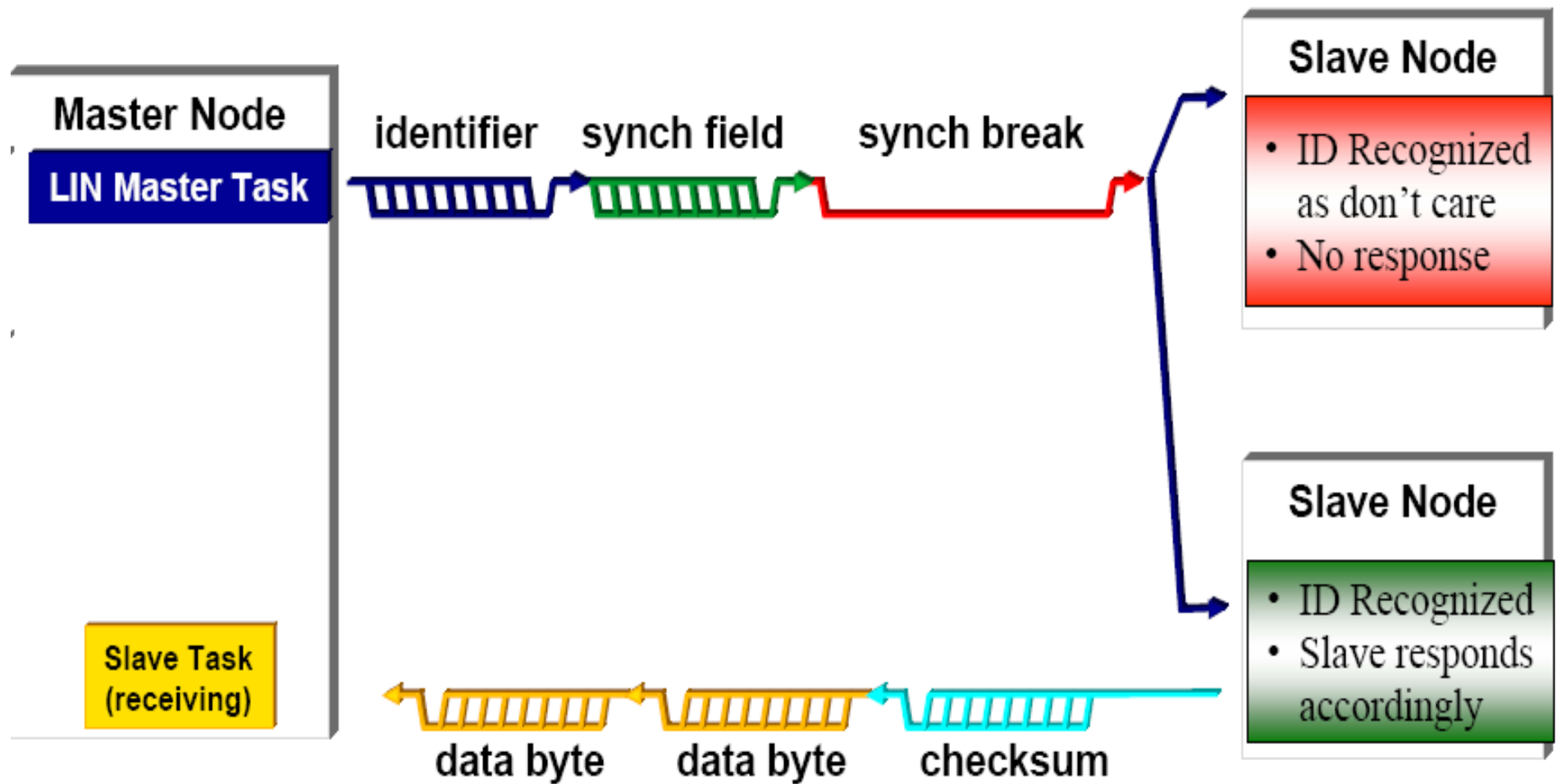
Master Sends Data to One Slave



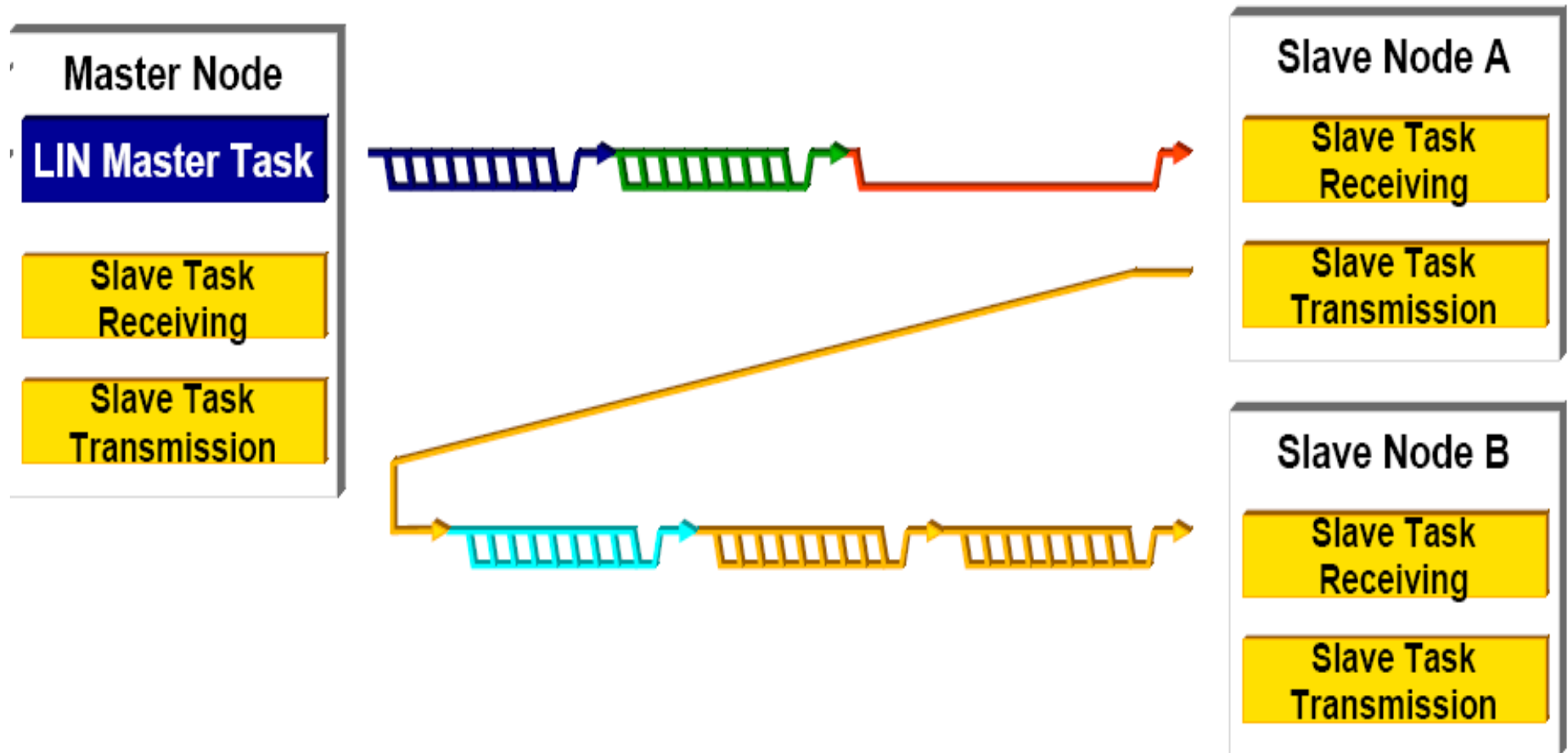
Master Sends Data to Many Slaves



Master Polls Slaves for Data



Data Transfer from Slave to Slave



Frame Types

- Unconditional frame
 - Carry signals and their frame identifiers are in the range 0 (zero) to 59 (0x3B)
- Diagnostic frames
 - The frame identifier is either 60 (0x3C), called master request frame, or 61 (0x3D), called slave response frame
- Reserved frames
 - Reserved frames shall not be used in a LIN 2.x cluster. Their frame identifier are 62 (0x3E) and 63 (0x3F)



Table of Valid Frame Identifiers

ID[0..5] Dec Hex		P0 = ID0⊕ID1⊕ID2⊕ID4	P1 = ¬ ID1⊕ID3⊕ID4⊕ID5	PID-Field P1 P0 5 4 3 2 1 0								PID-Field Dec Hex		
0	0x00	0	1	1	0	0	0	0	0	0	0	0	128	0x80
1	0x01	1	1	1	1	0	0	0	0	0	0	1	193	0xC1
2	0x02	1	0	0	1	0	0	0	0	1	0		66	0x42
3	0x03	0	0	0	0	0	0	0	0	1	1		3	0x03
4	0x04	1	1	1	1	0	0	0	1	0	0		196	0xC4
5	0x05	0	1	1	0	0	0	0	1	0	1		133	0x85
6	0x06	0	0	0	0	0	0	0	1	1	0		6	0x06
7	0x07	1	0	0	0	0	0	0	1	1	1		71	0x47
8	0x08	0	0	0	0	1	0	0	0	0			8	0x08
9	0x09	1	0	0	0	1	0	0	0	0	1		73	0x49
10	0x0A	1	1	1	0	0	1	0	1	0			202	0xCA
11	0x0B	0	1	0	0	0	1	0	1	1			139	0x8B
12	0x0C	1	0	0	0	1	1	0	0				76	0x4C
13	0x0D	0	0	0	0	1	1	0	1				13	0x0D



Table of Valid Frame Identifiers (Contd.)

ID[0..5] Dec Hex		P0 = ID0⊕ID1⊕ID2⊕ID4	P1 = ¬ ID1⊕ID3⊕ID4⊕ID5	PID-Field P1 P0 5 4 3 2 1 0								PID-Field Dec Hex	
14	0x0E	0	1	1	0	0	0	1	1	1	0	142	0x8E
15	0x0F	1	1	1	1	0	0	1	1	1	1	207	0xCF
16	0x10	1	0	0	1	0	0	0	0	0	0	80	0x50
17	0x11	0	0	0	1	0	0	0	0	1		17	0x11
18	0x12	0	1	0	1	0	0	1	0			146	0x92
19	0x13	1	1	0	1	0	0	1	1			211	0xD3
20	0x14	0	0	0	1	0	1	0	0			20	0x14
21	0x15	1	0	1	0	1	0	1	0	1		85	0x55
22	0x16	1	1	0	1	0	1	1	0			214	0xD6
23	0x17	0	1	0	1	0	1	1	1			151	0x97
24	0x18	1	1	0	1	1	0	0	0			216	0xD8
25	0x19	0	1	0	1	1	0	0	1			153	0x99
26	0x1A	0	0	0	1	1	0	1	0			26	0x1A
27	0x1B	1	0	1	1	0	1	1				91	0x5B
28	0x1C	0	1	1	1	0	0					156	0x9C
29	0x1D	1	1	0	1	1	1	0	1			221	0xDD



Table of Valid Frame Identifiers (Contd.)

ID[0..5] Dec Hex		P0 = ID0⊕ID1⊕ID2⊕ID4	P1 = ¬ ID1⊕ID3⊕ID4⊕ID5	PID-Field P1 P0 5 4 3 2 1 0								PID-Field Dec Hex	
30	0x1E	1	0	0	1	0	1	1	1	1	0	94	0x5E
31	0x1F	0	0	0	0	1	1	1	1	1	1	31	0x1F
32	0x20	0	0	1	0	0	0	0	0	0	0	32	0x20
33	0x21	1	0	1	0	0	0	0	0	1	1	97	0x61
34	0x22	1	1	1	0	0	0	1	0			226	0xE2
35	0x23	0	1	1	0	0	0	1	1			163	0xA3
36	0x24	1	0	1	0	0	1	0	0			100	0x64
37	0x25	0	0	1	0	0	1	0	1			37	0x25
38	0x26	0	1	1	0	0	1	1	0			166	0xA6
39	0x27	1	1	1	0	0	1	1	1			231	0xE7
40	0x28	0	1	1	0	1	0	0	0			168	0xA8
41	0x29	1	1	1	0	1	0	0	1			233	0xE9
42	0x2A	1	0	1	0	1	0	1	0			106	0x6A
43	0x2B	0	0	1	0	1	0	1	1			43	0x2B
44	0x2C	1	1	1	0	1	1	0	0			236	0xEC
45	0x2D	0	1	1	0	1	1	0	1			173	0xAD
46	0x2E	0	0	1	0	1	1	1	0			46	0x2E
47	0x2F	1	0	1	0	1	1	1	1			111	0x6F



Table of Valid Frame Identifiers (Contd.)

ID[0..5] Dec Hex		P0 = ID0⊕ID1⊕ID2⊕ID4	P1 = $\neg$ ID1⊕ID3⊕ID4⊕ID5	PID-Field P1 P0 5 4 3 2 1 0								PID-Field Dec Hex	
48	0x30	1	1	1	1	1	1	0	0	0	0	240	0xF0
49	0x31	0	1	1	1	1	0	0	0	0	1	177	0xB1
50	0x32	0	0	1	1	0	0	1	0	0	0	50	0x32
51	0x33	1	0	1	1	0	0	1	1	0	0	115	0x73
52	0x34	0	1	1	1	0	1	0	0	0	0	180	0xB4
53	0x35	1	1	1	1	0	1	0	1	0	1	245	0xF5
54	0x36	1	0	1	1	0	1	1	0	0	0	118	0x76
55	0x37	0	0	1	1	0	1	1	1	1	1	55	0x37
56	0x38	1	0	1	1	1	0	0	0	0	0	120	0x78
57	0x39	0	0	1	1	1	0	0	1	0	1	57	0x39
58	0x3A	0	1	1	1	1	0	1	0	0	0	186	0xBA
59	0x3B	1	1	1	1	1	0	1	1	1	1	251	0xFB
60 ^a	0x3C	0	0	1	1	1	1	0	0	0	0	60	0x3C
61 ^b	0x3D	1	0	1	1	1	1	0	1	0	1	125	0x7D
62 ^c	0x3E	1	1	1	1	1	1	1	1	0	0	254	0xFE
63 ^d	0x3F	0	1	1	1	1	1	1	1	1	1	191	0xBF

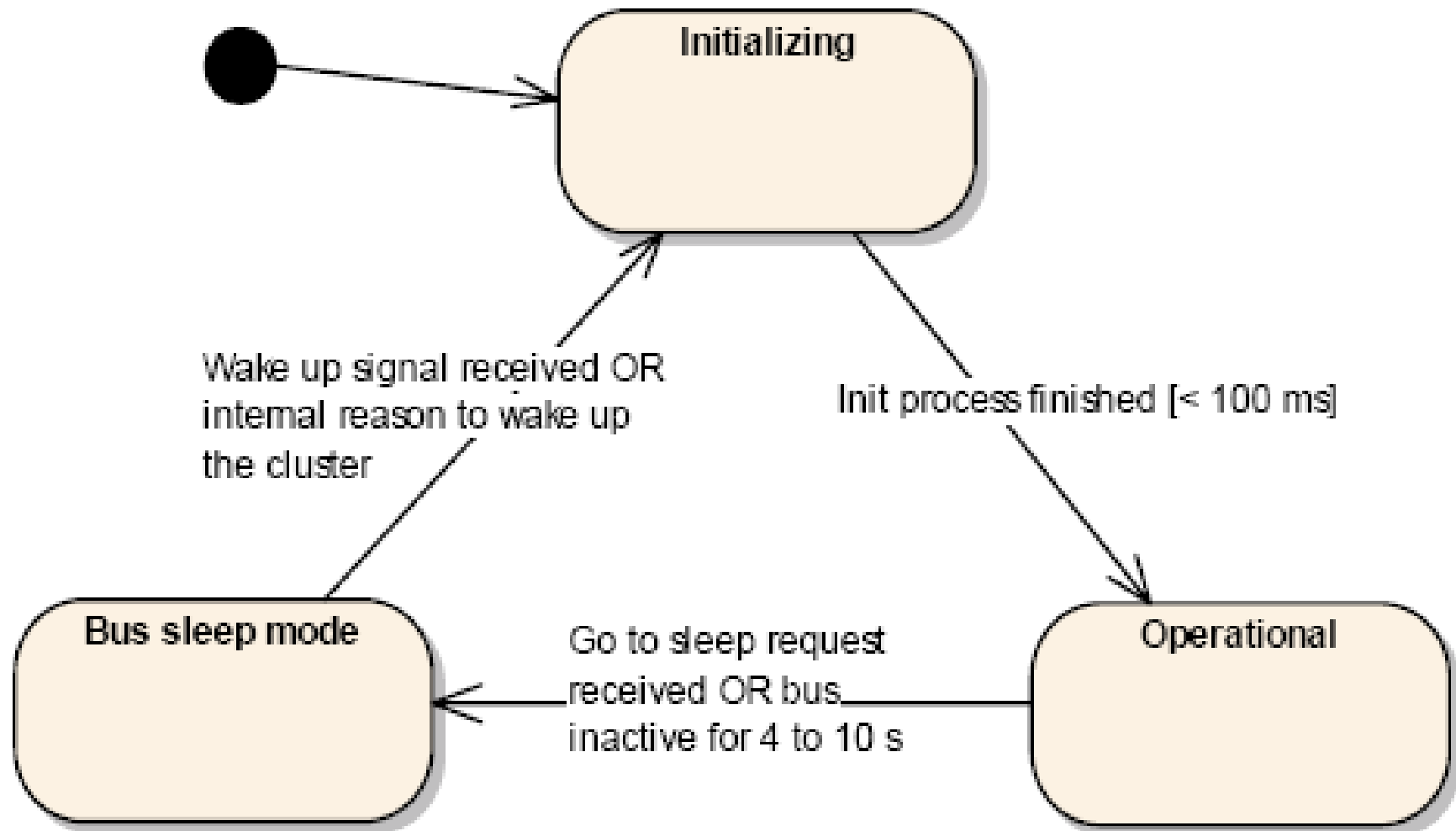


Schedule Tables

- A key property of the LIN protocol
- Guarantee the periodicity of signals
- Assures that the bus will never be overloaded
- Deterministic behavior initiated by the master task
- All frames relevant in a mode of operation are given enough time to be transferred



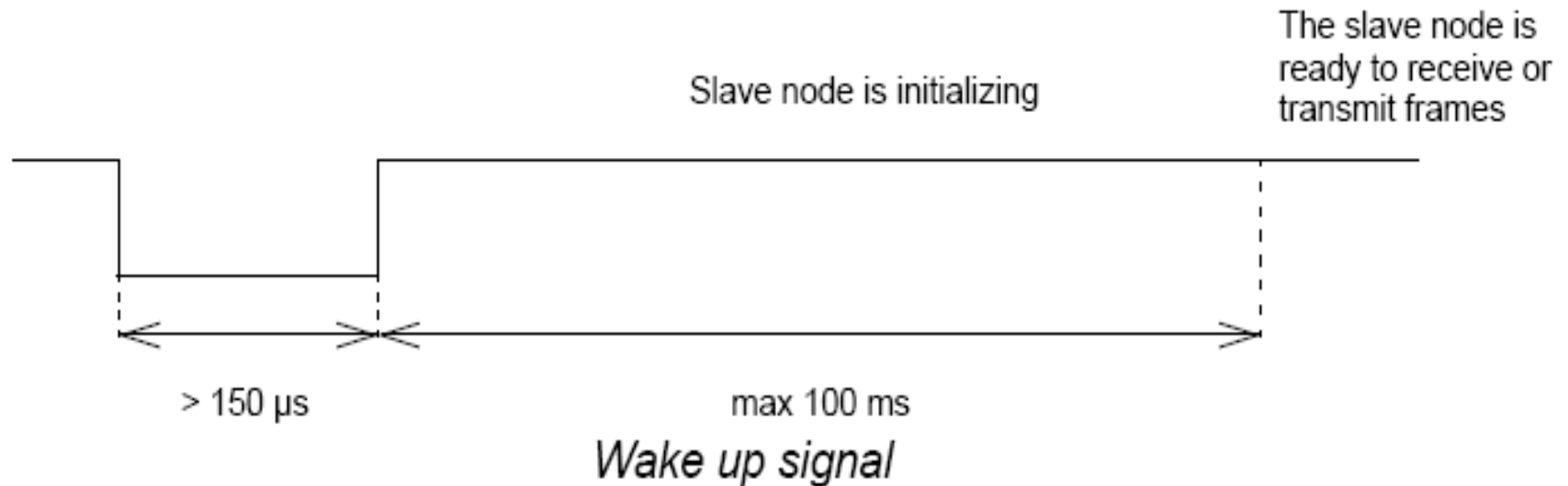
Network Management of LIN Cluster



Slave node communication state diagram

Wake Up Mode

- Any node in a sleeping LIN cluster may request a wake up, by transmitting a wake up signal
- The wake up signal is issued by forcing the bus to the dominant state for 250 μ s to 5 ms
- Every slave node (connected to power) shall detect the wake up signal (a dominant pulse longer than 150 μ s) and be ready to listen to bus commands within 100 ms, measured from the ending edge of the dominant pulse



Sleep Mode

- The master sets the cluster into bus sleep mode by transmitting a go to sleep command
- The slave node application may still be active after the go to sleep command has been received. This behavior is application specific
- Master node transmits the go to sleep command
- The slave node automatically enters bus sleep mode earliest 4 s and latest 10 s of bus inactivity ()

data1	data2	data3	data4	data5	data6	data7	data8
0	0xFF	0xFF	0xFF	0xFF	0xFF	0xFF	0xFF

Go to sleep command



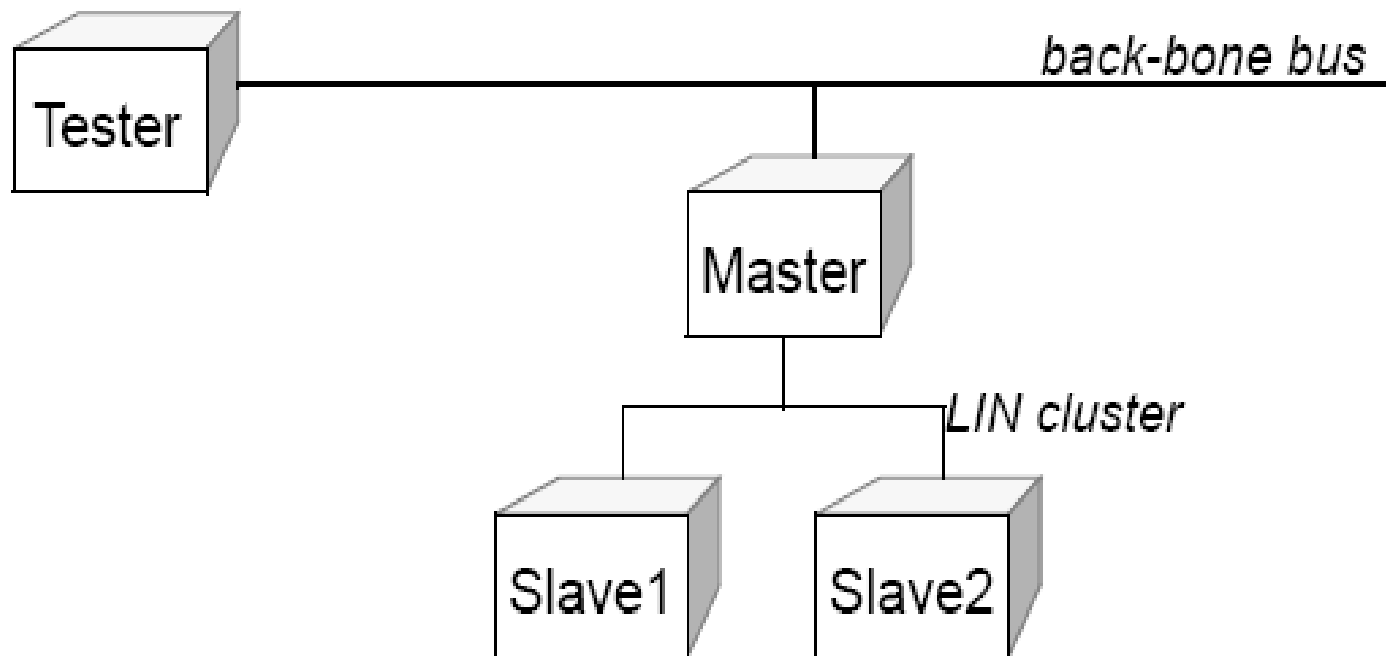
Network Management of LIN Cluster

- Refers mainly to cluster wake up and go to sleep
- Behavior model for communication of a slave node:
 - **Initializing:**
 - This state is entered after reset and after wakeup
 - The slave node makes necessary initialization and then enters the Operational state
 - **Operational:**
 - The LIN protocol behavior (transmitting and receiving frames) only applies to the Operational state
 - **Bus sleep mode:**
 - The level on the bus is set to recessive
 - Only the wake up signal may be transmitted on the cluster



LIN Transport Layer

- Transport layer is used when diagnostics are performed on the back-bone bus (e.g. CAN) and where the system builder wants to use the same diagnostic capabilities on the LIN sub-bus clusters



Typical system setup for a LIN cluster using the transport layer.



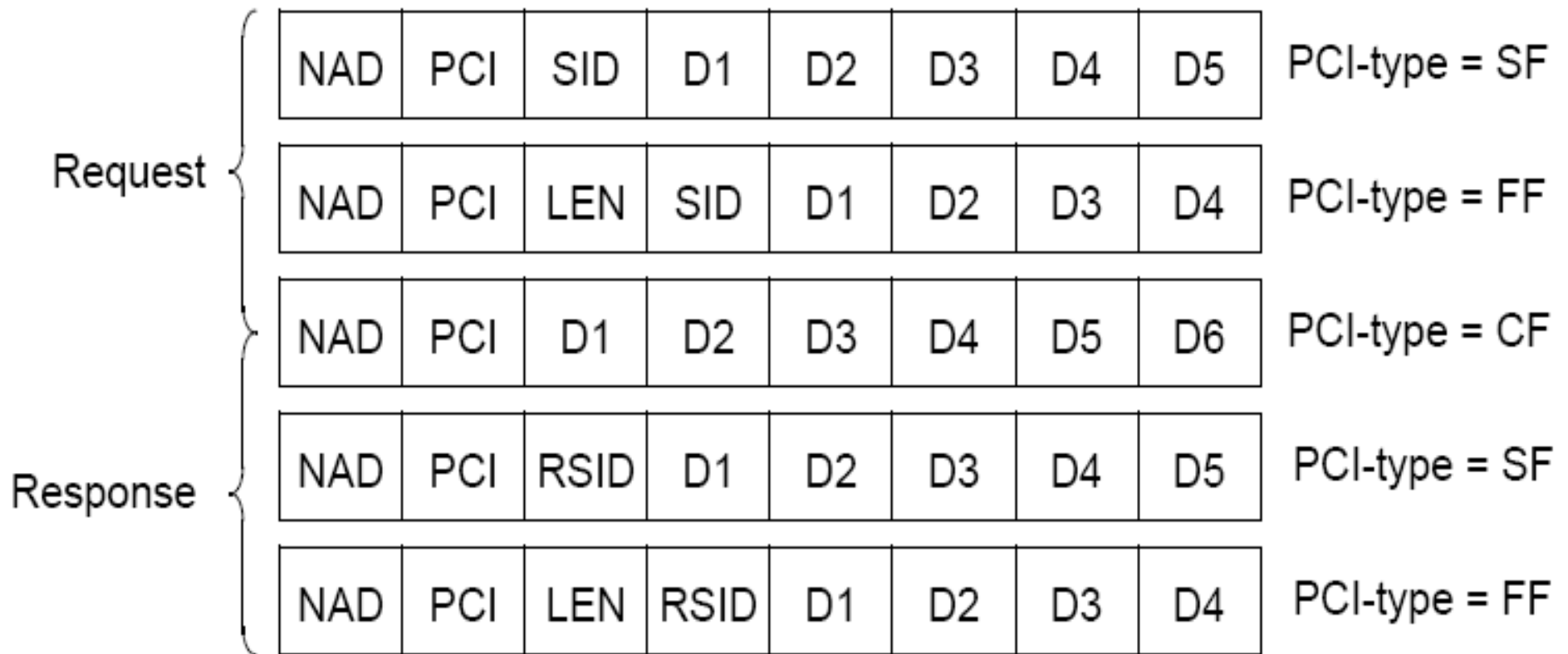
Packet Data Unit

- The units that are transported in a transport layer frame are called PDU (Packet Data Unit)
- Messages issued by the client (tester, master node) are called requests
- Messages issued by the server (master node, slave node) are called responses
- Requests are always sent in master request frames and responses are always sent in slave response frames



PDU Structure

- The left byte called Node Address (NAD) is sent first and the right byte (D4, D5 or D6) is sent last



PDUs supported by the LIN transport layer.



Node Address (NAD)

- Only slave nodes have an node address
- NAD is the address of the slave node being addressed in a request
- The address is used to indicate the source of a response

NAD value	Description
0	Reserved for go to sleep command
1 - 125 (0x7D)	Slave node addresses (NAD)
126 (0x7E)	Functional node address (functional NAD), only used for diagnostics (using the transport layer)
127 (0x7F)	Slave node address broadcast (broadcast NAD)
128 (0x80) - 255 (0xFF)	Free usage. Diagnostic frames with the first byte in the range 128 (0x80) to 255 (0xFF) are allocated for free usage since the LIN 1.2 standard.



Protocol Control Information (PCI)

- Contains the transport layer flow control information
- The PCI type Single Frame (SF) indicates that the transported message fits into the single PDU (maximum 5 data bytes)
- The PCI type First Frame (FF) is used to indicate the start of a multi PDU message, the following frames are of CF type

Type	PCI type				Additional information			
	B7	B6	B5	B4	B3	B2	B1	B0
SF	0	0	0	0	Length			
FF	0	0	0	1	Length			
CF	0	0	1	0	Frame counter			

Structure of the PCI byte

Protocol Control Information (PCI)

- A multi-PDU message is continued with a number of Consecutive Frames (CF).
- The first CF frame of a message is numbered 1, the second 2 and so on.
- If more than 15 CF PDUs are needed to transport the complete message, the frame counter wraps around and continues with 0, 1,...
- LEN
 - A LEN byte is only used in FF; it contains the eight least significant bits of the message length. Thus, the maximum length of a message is 4095 (0xFFFF) bytes.



Service Identifier (SID)

- SID

- The Service Identifier (SID) specifies the request that shall be performed by the slave node addressed
- 0 to 0xAF and 0xB8 to 0xFE are used for diagnostics
- 0xB0 to 0xB7 are used for node configuration (Defined by vehicle manufacturer)
- The Response Service Identifier (RSID) specifies the contents of the response. The RSID for a positive response is always SID + 0x40



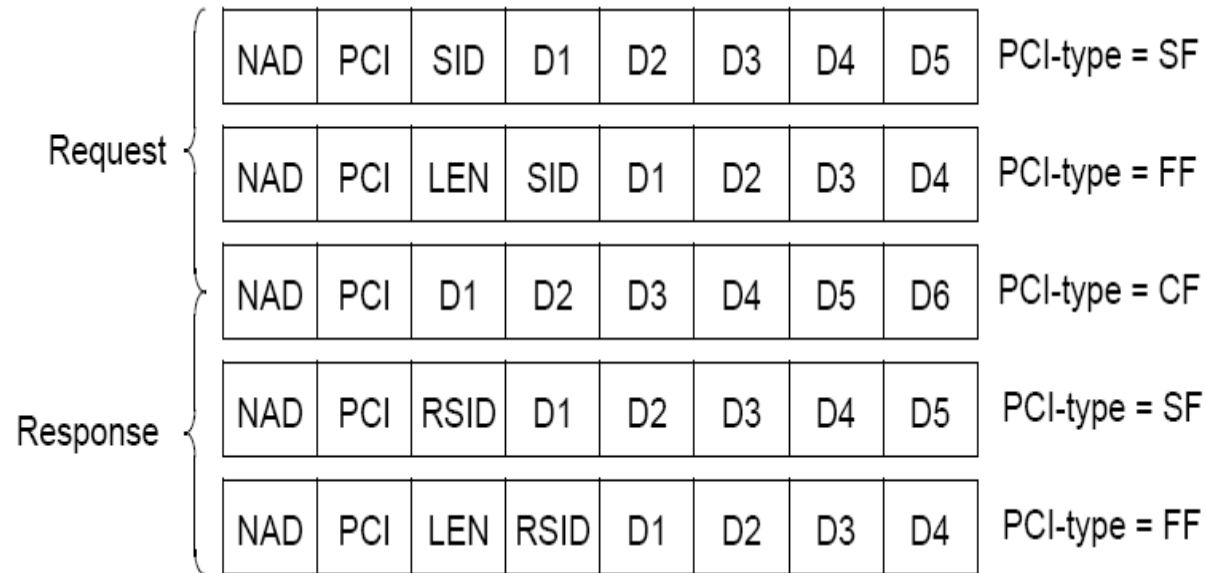
Service Identifier (SID)

SID	Service	type
0 - 0xAF	reserved	reserved
0xB0	Assign NAD	Optional
0xB1	Assign frame identifier	Obsolete
0xB2	Read by Identifier	Mandatory
0xB3	Conditional Change NAD	Optional
0xB4	Data Dump	Optional
0xB5	Assign NAD via SNPD	Reserved by SNPD
0xB6	Save Configuration	Optional
0xB7	Assign frame identifier range	Mandatory
0xB8 - 0xFF	reserved	reserved



Data Bytes

- D1 to D6
 - Up to six data bytes in a single PDU
 - The interpretation of the data bytes depends on the SID or RSID
 - If a PDU is not completely filled (applies to CF and SF PDUs only) the unused bytes shall be filled with ones, i.e. their value shall be 255 (0xFF)



Frame Transmission

- Single Frame Transmission
 - Transmission of messages up to six bytes (including SID) shall be performed via transmission of a single frame PDU (SF)
- Multiple Frame Transmission
 - Transmission of messages with more than six bytes (including SID) up to a maximum of 4095 bytes is performed via segmentation and transmission of multiple PDUs
 - A segmented transmission starts with a First Frame PDU (FF) and continues with multiple Consecutive Frame PDUs (CF)

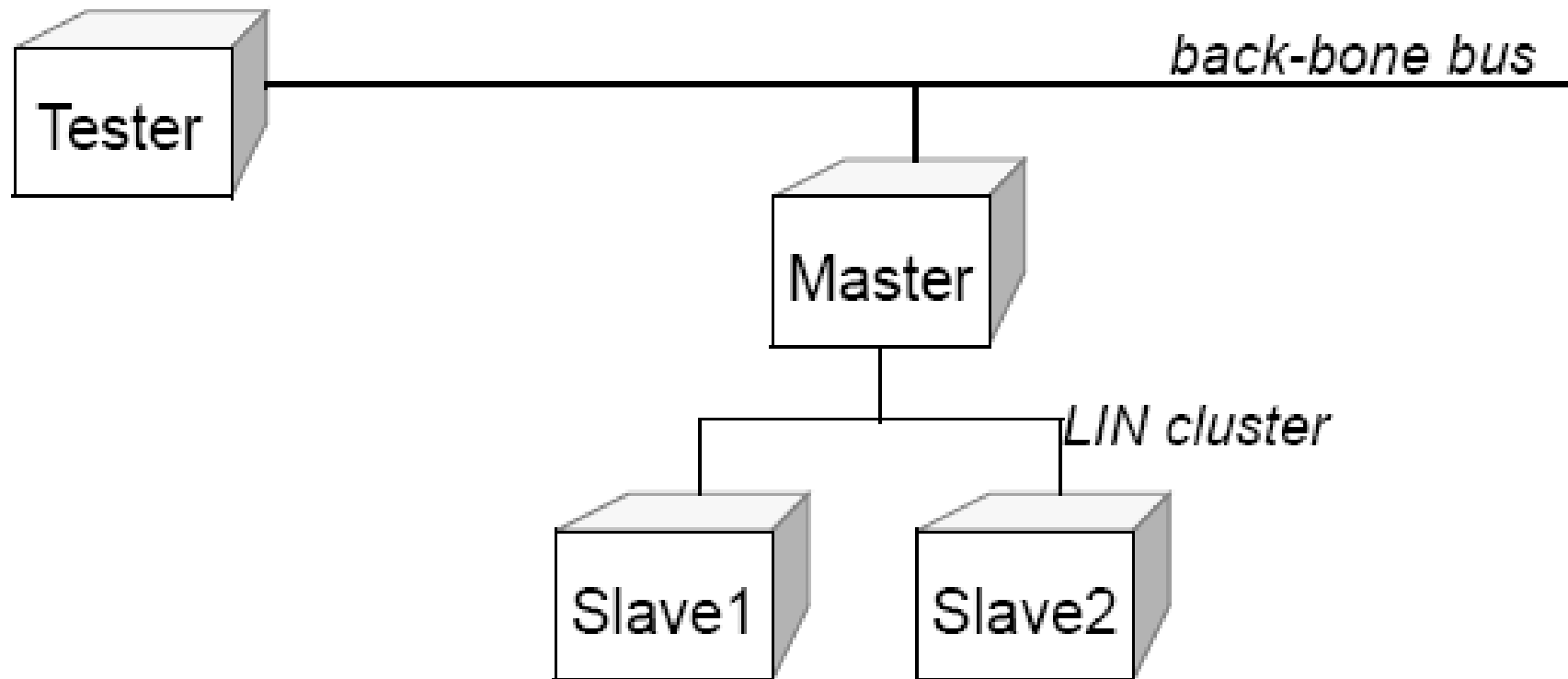


Error Handling

- LIN frames are ignored in the following situations:
 - A Single Frame PDU (SF) with a length value greater than six bytes
 - First Frame PDU (FF) with a length value less than seven bytes
 - First Frame PDU (FF) with a length value greater than the maximum available receive buffer size of the slave node
 - PDUs with unexpected PCI types from any node shall be ignored

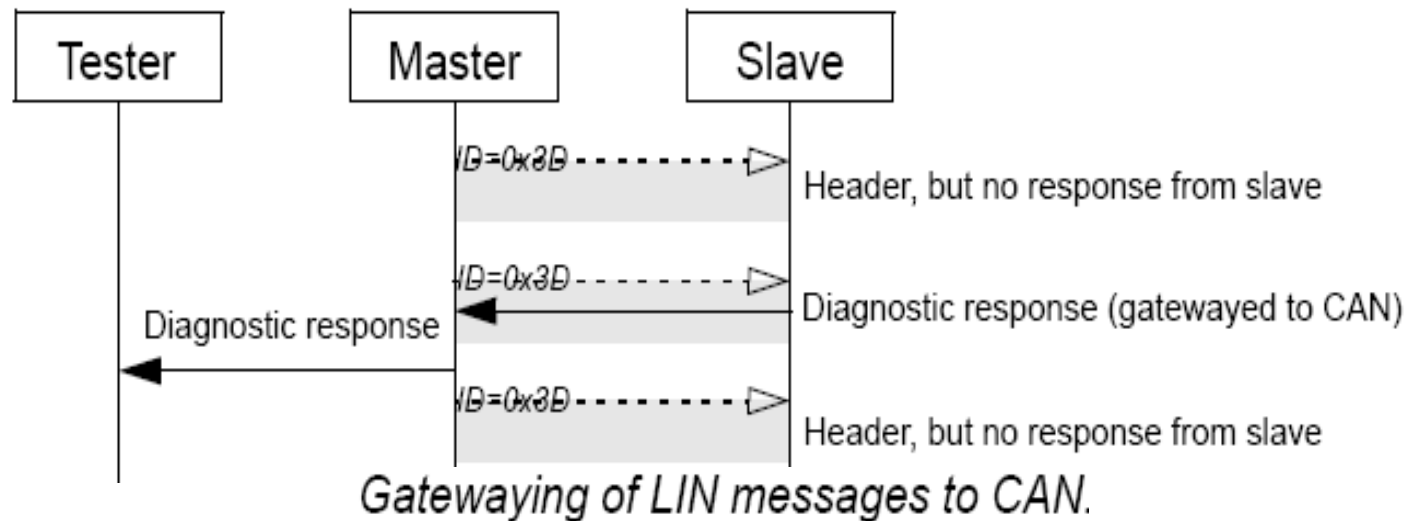
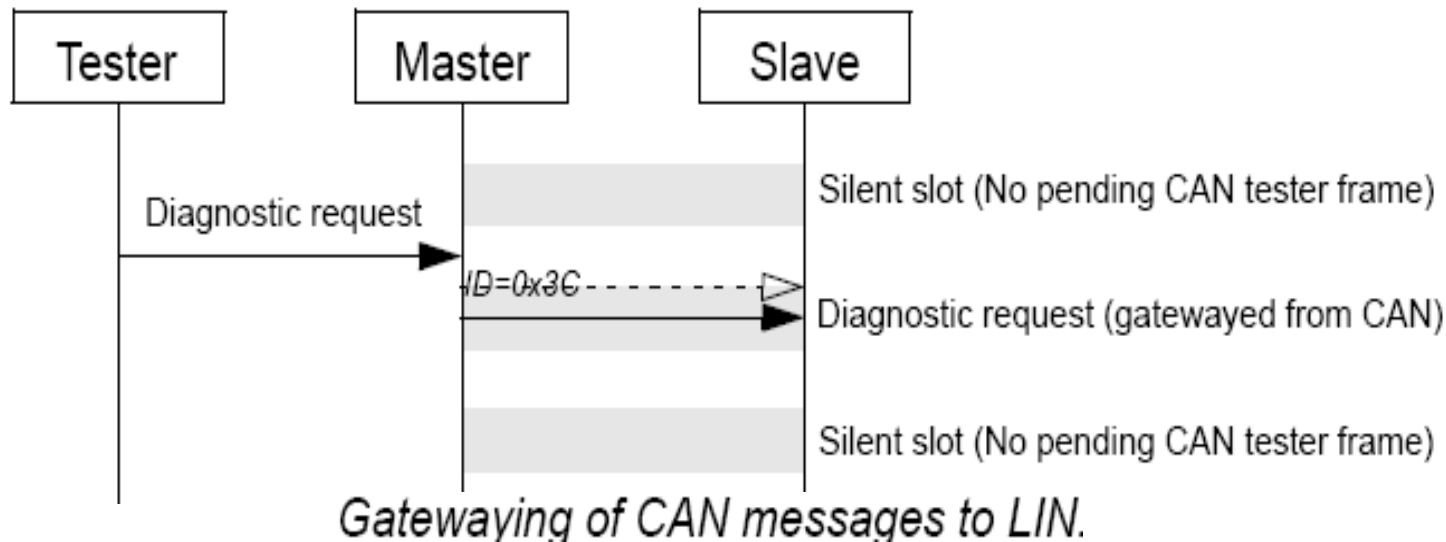


LIN Diagnostic

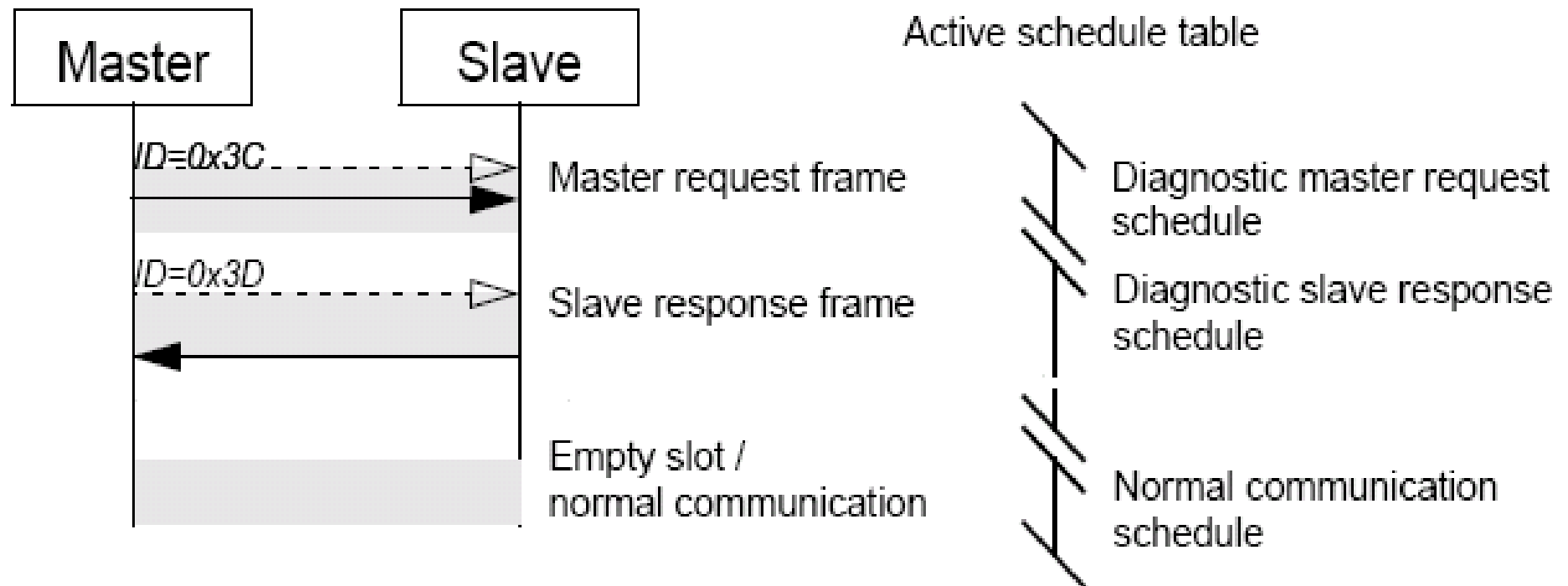


Typical system setup for a LIN cluster using the transport layer.

LIN Diagnostic



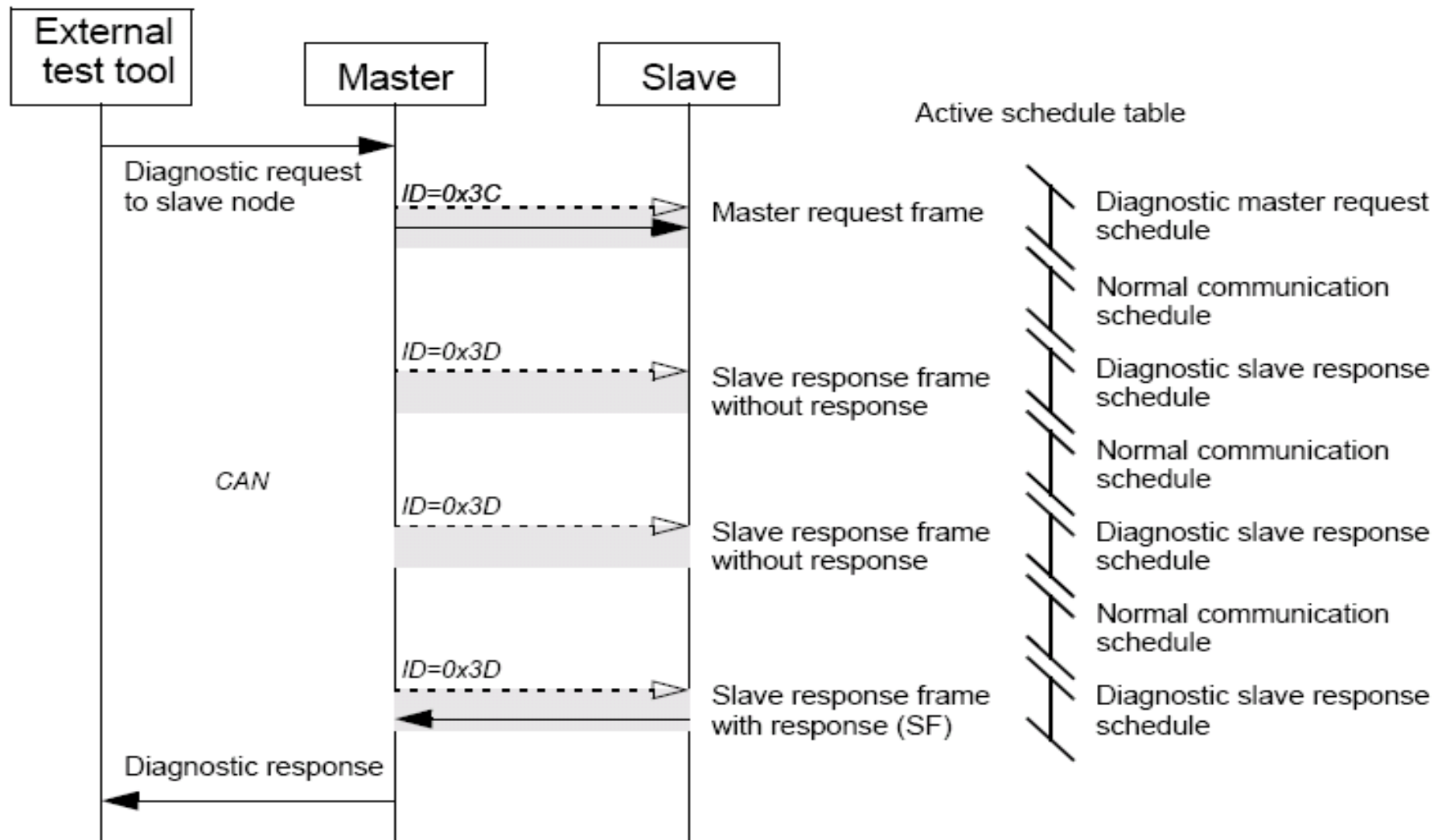
Diagnostic Scheduling in LIN Master



Single execution of request – response frames



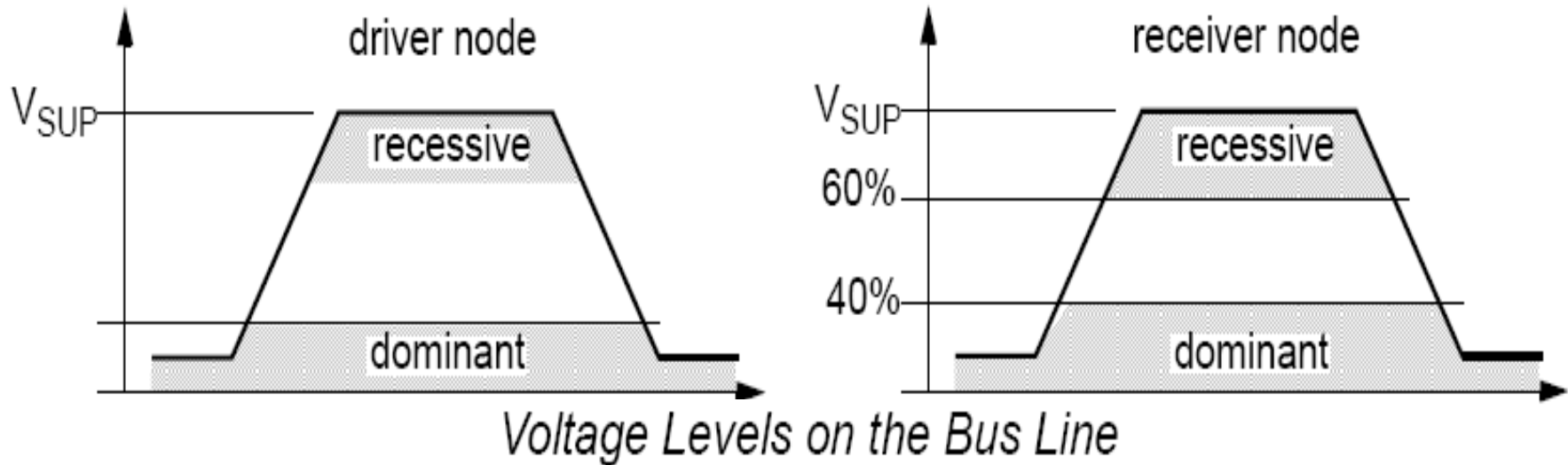
Diagnostic Scheduling in LIN Master



Continuous execution of request – response frames

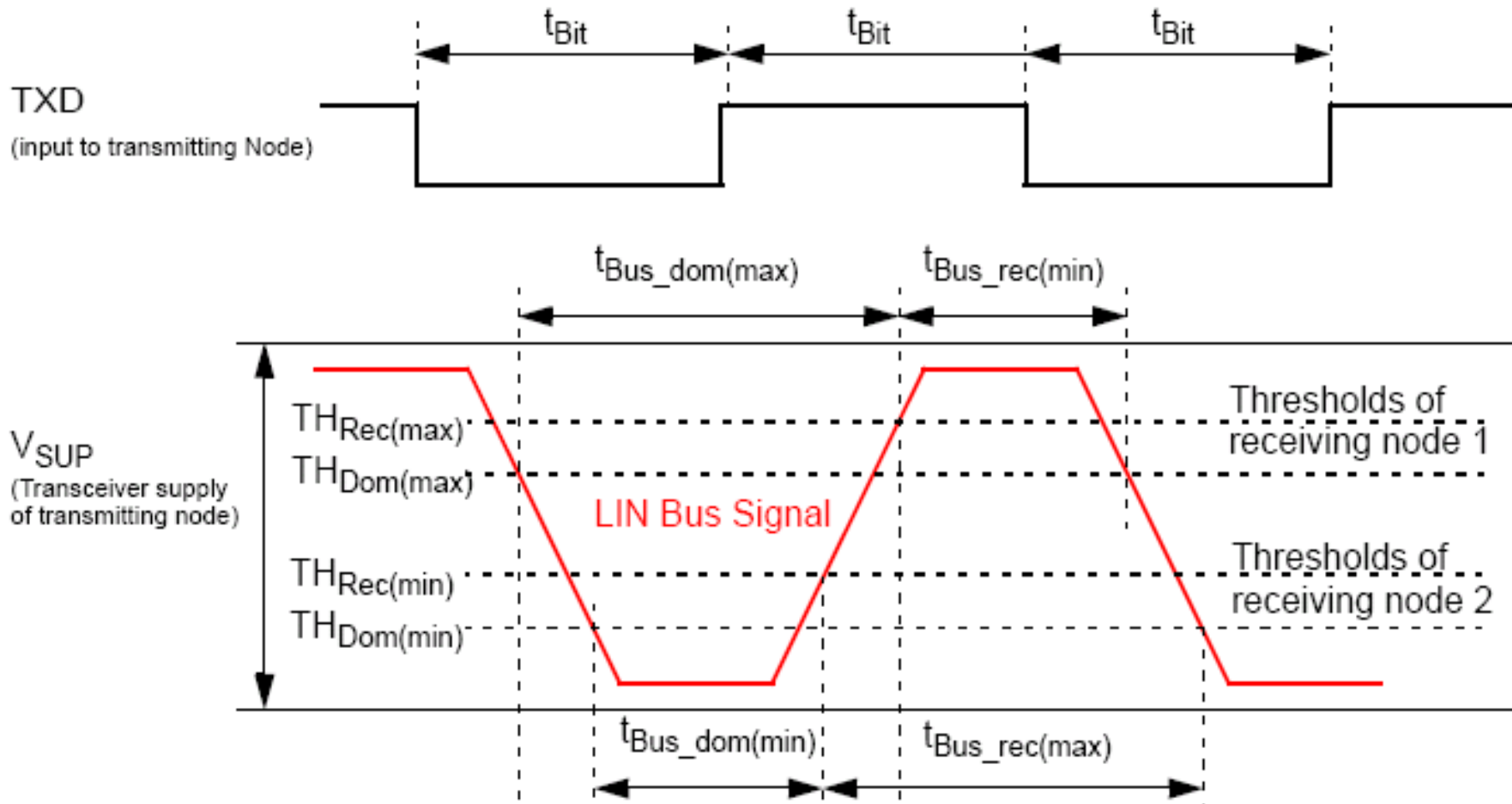


LIN-PHY Signal Specification



- Expected voltage level (dominant or recessive) at the bit sampling time of the receiver

LIN-PHY Bit Time



LIN vs CAN

Parameter	LIN	CAN
Medium Access Control	Single Master	Multi Master
Maximum Supported Data Rate	Up to 20 Kbps	Up to 1 Mbps
Length of Identifier	6 bits	11/29 bits
Maximum Data Bytes	8 Bytes	8 Bytes
Error Detection Field	Parity (protected identifier) 8 bit checksum (frame)	15 bit CRC
Quartz/Ceramic Resonator	Master Only	All Nodes
Typical Size of Network	2-16 Nodes	4-20 Nodes



Summary

- The primary and original purpose of LIN is to provide a 'sub-bus' for CAN, with reduced functionality and lower costs
- LIN is used for applications where
 - bit rate requirements of the network are low
 - Reliability and robustness found in CAN are not required
- All messages sent on the LIN-bus are byte-encoded according to the UART-protocol
- A frame comprises of
 - Header and
 - Response



Summary

- There are three types of frames
 - Unconditional
 - Diagnostic
 - Reserved
- LIN transport layer is used when diagnostics are performed on the back-bone bus
- LIN node can be in three states
 - Initializing
 - Operational
 - Bus sleep mode

